September 18, 2026

ECEN 474/704
Lab 3: Prelab
Anonymous contributor

1. **Minimum transistor dimensions.** What are the absolute minimum dimensions of a transistor? Explain your reasoning, including the minimum contact size. Do we use this minimum size or a slightly larger size for convenience?

   Solution.

   (a) Minimum channel dimensions. For the low-voltage MOS devices used in this lab, the CMHV7SF design manual specifies the following drawn dimensions:

   $$\boxed{L_{\text{min}} = 0.180\,\mu\text{m}, \qquad W_{\text{min,channel}} = 0.220\,\mu\text{m}.}$$

   These are the channel dimensions, rather than the dimensions of the entire contacted transistor layout.

   (b) Contact constraint. A source/drain contact (CA) is $0.200\,\mu\text{m}\times0.200\,\mu\text{m}$, and the minimum active-region (RX) enclosure is $0.100\,\mu\text{m}$ on each side. Thus the diffusion width at a contact must be at least

   $$W_{\text{contact region}} = 0.100 + 0.200 + 0.100 = 0.400\,\mu\text{m}.$$

   If the source, channel, and drain use one constant-width rectangular diffusion strip, the minimum contacted dimensions are therefore

   $$\boxed{W_{\text{min,rectangular}} = 0.400\,\mu\text{m}, \qquad L_{\text{min}} = 0.180\,\mu\text{m}.}$$

   A $0.220\,\mu\text{m}$ channel remains possible with a dogbone-shaped diffusion region: the source/drain contact regions are widened to at least $0.400\,\mu\text{m}$, while the channel stays narrower. The diffusion transitions must also satisfy the gate-to-RX-corner spacing rules. A $0.200\,\mu\text{m}$ contact cannot simply be placed inside a constant-width $0.220\,\mu\text{m}$ strip.

   (c) Practical choice. For convenient contact placement, repeated fingers, and routing, this lab uses a slightly larger finger width:

   $$\boxed{W_f = 0.500\,\mu\text{m}, \qquad L = 0.180\,\mu\text{m}.}$$

   The width is larger than the minimum contacted rectangular width, while the channel length remains at its minimum. The $0.500\,\mu\text{m}$ width is a convenient design choice, not the absolute minimum specified by the PDK.

   Rule references: CMHV7SF Design Manual, applicable to V1.3.1.0, Table 24, p. 67, Rules 1–2 and 10–11; Table 26, p. 77, Rules 200 and 204. The dogbone transition is also subject to Rule 114 on p. 69.

2. **Common-centroid current mirror.** Draw a common-centroid layout of a simple current mirror with equal transistors of $L = 180\,\text{nm}$ and $W = 12\,\mu\text{m}$. Use a finger width of 500 nm, show the drain, source, gate, and bulk connections, include dummy transistors, and provide a floor plan.

Solution.

(a) Finger count and floor plan. The number of fingers in each branch is

$$n_f = \frac{W}{W_f} = \frac{12}{0.50} = 24.$$

There are 48 functional fingers in total. Choose four rows of 12 functional fingers, with one dummy at each end of every row:

$$\boxed{4 \times 12 \text{ functional fingers + 8 dummies.}}$$

Let A denote a reference-branch finger, B an output-branch finger, and X a dummy. The floor plan is

X	A	B	B	A	A	B	B	A	A	B	B	A	X
X	B	A	A	B	B	A	A	B	B	A	A	B	X
X	B	A	A	B	B	A	A	B	B	A	A	B	X
X	A	B	B	A	A	B	B	A	A	B	B	A	X

Each row contains six A and six B fingers, giving 24 of each overall. With equal finger pitch and symmetric row placement, the two groups have the same centroid. Each functional finger has $W_f/L = 0.50/0.18\,\mu\text{m}$; dummies do not contribute to the specified effective transistor width.

(b) Electrical connections. For an NMOS current mirror, use the following nets:

Finger	D	G	S	Bulk
A	I_{IN}	I_{IN}	V_{SS}	V_{SS}
B	I_{OUT}	I_{IN}	V_{SS}	V_{SS}
X	Adjacent D_1 or D_2	V_{SS}	V_{SS}	V_{SS}

The reference branch is diode-connected: $G = D_1 = I_{\text{IN}}$. All functional gates share this node, while $D_2 = I_{\text{OUT}}$. Each dummy has $V_{GS} = 0$ and its gate is isolated from the functional gate bus. Substrate taps connect the bulk to V_{SS}.

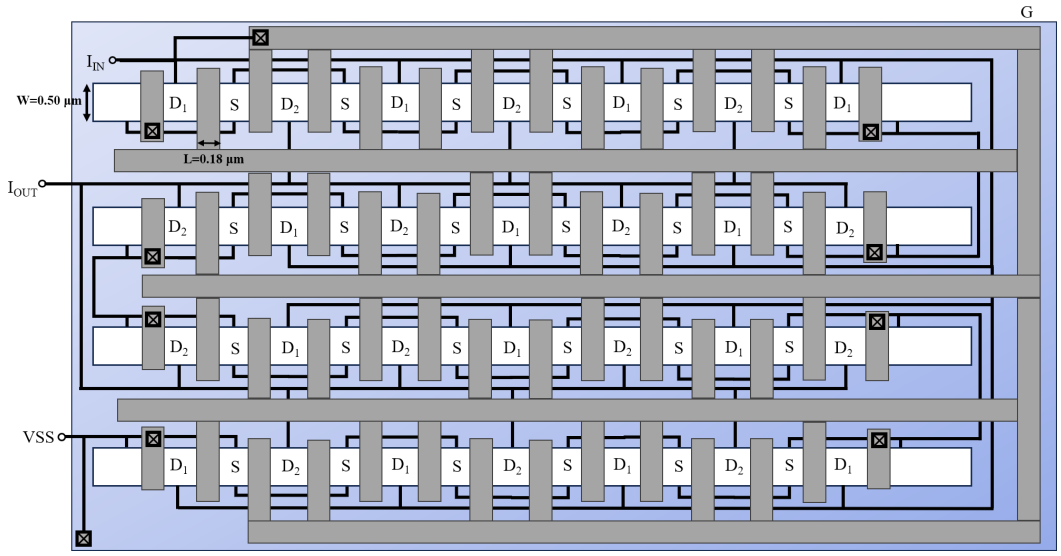


Figure 1: Common-centroid current-mirror layout with four rows, twelve functional fingers per row, and eight edge dummies.

(c) Matching considerations. Use equal finger dimensions and regular spacing, share diffusion where appropriate, and balance the drain routing, contact counts, and parasitic loading between the branches. The outer two rows and inner two rows exchange A and B, balancing the drain regions at the row ends. Common-centroid placement suppresses first-order spatial-gradient mismatch; it does not eliminate random mismatch.

3. **Common-centroid MOM capacitor array.** Draw a common-centroid MOM capacitor array with $C_1 : C_2 = 1 : 1.4$, using eight unit capacitors ($I_1 = I_2 = 4$) and one non-unit capacitor. Each unit measures $10.56\,\mu\text{m} \times 10.56\,\mu\text{m}$. Determine the non-unit dimensions, show the interconnections, and include dummy capacitors and suitable matching techniques.

Solution.

(a) Required non-unit capacitance. Let C_u be the unit capacitance and $C_{\text{nu}} = NC_u$. Then

$$C_1 = 4C_u, \qquad C_2 = (4+N)C_u, \qquad \frac{4}{4+N} = \frac{1}{1.4}.$$

Therefore,

$$\boxed{N = 1.6, \qquad C_{\text{nu}} = 1.6C_u, \qquad C_2 = 5.6C_u.}$$

(b) Non-unit capacitor dimensions.

Using $L_0 = 10.56\,\mu\text{m}$ and the lab manual's area/perimeter matching formulas,

$$\begin{aligned} L_{\text{nu}} &= L_0\left[N + \sqrt{N(N-1)}\right] \\ &= 10.56\left[1.6 + \sqrt{1.6(0.6)}\right] = 27.24264467\,\mu\text{m}, \\ W_{\text{nu}} &= \frac{NL_0^2}{L_{\text{nu}}} = \frac{1.6(10.56)^2}{27.24264467} = 6.54935533\,\mu\text{m}. \end{aligned}$$

Thus,

$$\boxed{W_{\text{nu}} \times L_{\text{nu}} \simeq 6.549\,\mu\text{m} \times 27.243\,\mu\text{m}.}$$

The area and perimeter scale by the same factor:

$$\begin{aligned} A_u &= 111.5136\,\mu\text{m}^2, & P_u &= 42.24\,\mu\text{m}, \\ A_{\text{nu}} &= 178.42176\,\mu\text{m}^2 = 1.6A_u, & P_{\text{nu}} &= 67.584\,\mu\text{m} = 1.6P_u. \end{aligned}$$

Consequently, $P_{\text{nu}}/A_{\text{nu}} = P_u/A_u$. These are the nominal dimensions from the lab's matching construction; the implemented MOM capacitance also depends on the PDK model and available geometry increments.

(c) Common-centroid floor plan. Place the four A units of C_1 at the corners of the inner array, the four B units of C_2 at the side centers, and C_{nu} at the center. Surround the array with unit-size dummies X:

$$\begin{array}{ccccc} X & X & X & X & X \\ X & A_1 & B_1 & A_2 & X \\ X & B_2 & C_{\text{nu}} & B_3 & X \\ X & A_3 & B_4 & A_4 & X \\ X & X & X & X & X \end{array}$$

This gives four A units, four B units, one non-unit capacitor, and sixteen dummies. The table denotes placement, not equal-size physical cells: C_{nu} is a long rectangle, so the spacing must accommodate it symmetrically. Both unit groups have their centroids at the center, and adding the centered C_{nu} preserves the capacitance-weighted centroid of C_2.

(d) Interconnections and matching. Connect A_1–A_4 in parallel and connect B_1–B_4 and C_{nu} in parallel. In particular, C_{nu} belongs only to C_2. All backplate terminals connect to V_{SS}.

Use identical orientation and local routing for the unit capacitors, multiple contacts where appropriate, and balanced routing parasitics. Following Fig. 3-10 of the lab manual, short the two signal terminals of each dummy together as an individual floating node; its backplate remains connected to V_{SS}.

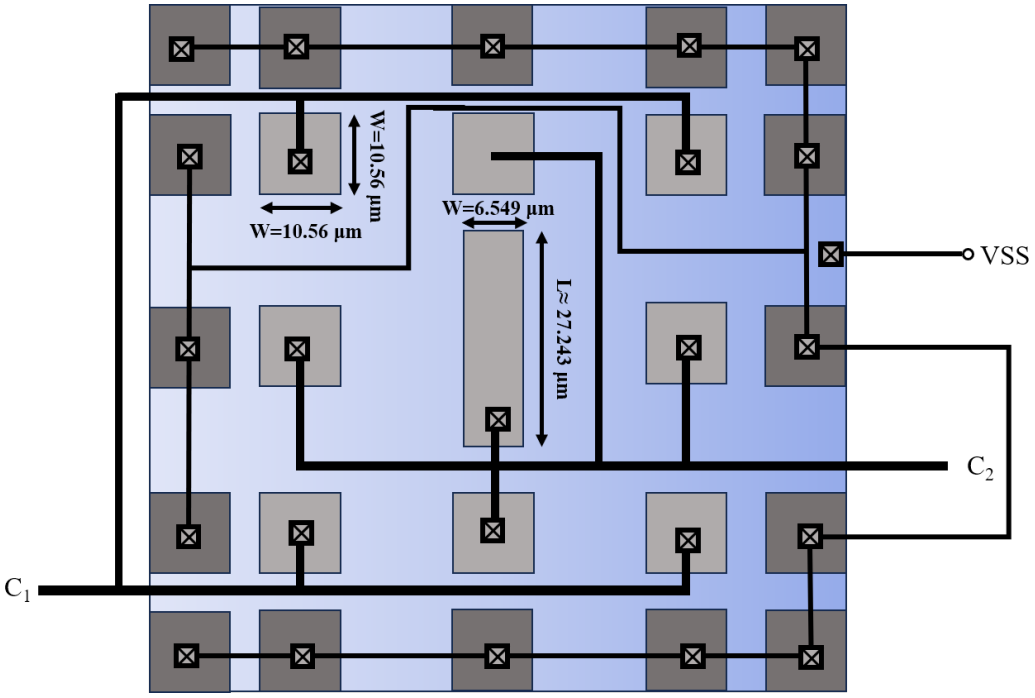


Figure 2: Common-centroid MOM capacitor array. The four C_1 units are connected in parallel; the four C_2 units and the centered non-unit capacitor are connected in parallel.

4. **Interdigitized matched polysilicon resistors.** Draw two matched `oprppresx` resistors of 15 kΩ each, using four series segments per resistor. Account for contact resistance, determine the segment length and width, and use an interdigitized layout. The combined component area of the two resistors must be between 30 and 80 μm^2.

Solution.

(a) Segment dimensions including contact resistance. Each of the four series segments must have nominal resistance

$$R_{\text{seg}} = \frac{15}{4} = 3.75\,\text{k}\Omega.$$

From Table 3-1 of the lab manual,

$$R_s = 0.165\,\text{k}\Omega/\square, \qquad R_{\text{con}} = 0.015\,\text{k}\Omega \cdot \mu\text{m}.$$

The nominal resistance of each contacted segment is

$$R_{\text{seg}} = R_s \frac{L}{W} + \frac{2R_{\text{con}}}{W}.$$

Choose $W = 0.50\,\mu$m. Measuring L in μm and resistance in kΩ gives

$$3.75 = 0.165\frac{L}{0.50} + \frac{2(0.015)}{0.50} = 0.33L + 0.06.$$

Therefore,

$$\boxed{L = 11.1818\,\mu\text{m}, \qquad W = 0.50\,\mu\text{m}.}$$

Each segment contributes 3.69 kΩ from its body and 0.06 kΩ from its two contacts. Hence

$$\boxed{R_1 = R_2 = 4(3.69 + 0.06) = 15.00\,\text{k}\Omega.}$$

The length L is the effective resistor-body length defined by the OP mask, rather than the full strip length including the contact ends.

(b) Area check. The combined body area of the eight functional segments is

$$A_{\text{active}} = 8WL = 8(0.50)(11.1818) \simeq \boxed{44.73\,\mu\text{m}^2},$$

which lies between 30 and 80 μm^2. Including two identical dummy bodies gives approximately 55.91 μm^2. The overall layout footprint is larger because contacts, routing, and spacing require additional area.

(c) Placement and series connections. Place ten parallel strips with equal orientation and spacing:

$$X \quad A_1 \quad B_1 \quad A_2 \quad B_2 \quad A_3 \quad B_3 \quad A_4 \quad B_4 \quad X.$$

The two groups are spatially interdigitized and electrically connected as

$$\text{R1} \longrightarrow A_1 \longrightarrow A_2 \longrightarrow A_3 \longrightarrow A_4 \longrightarrow \text{R1},$$
$$\text{R2} \longrightarrow B_1 \longrightarrow B_2 \longrightarrow B_3 \longrightarrow B_4 \longrightarrow \text{R2}.$$

For a serpentine connection, join the bottom ends of segments 1 and 2, the top ends of segments 2 and 3, and the bottom ends of segments 3 and 4 within each group. The top ends of segments 1 and 4 are the external terminals. This is an interdigitized arrangement, as requested; the single-row $ABABABAB$ pattern is not common-centroid.

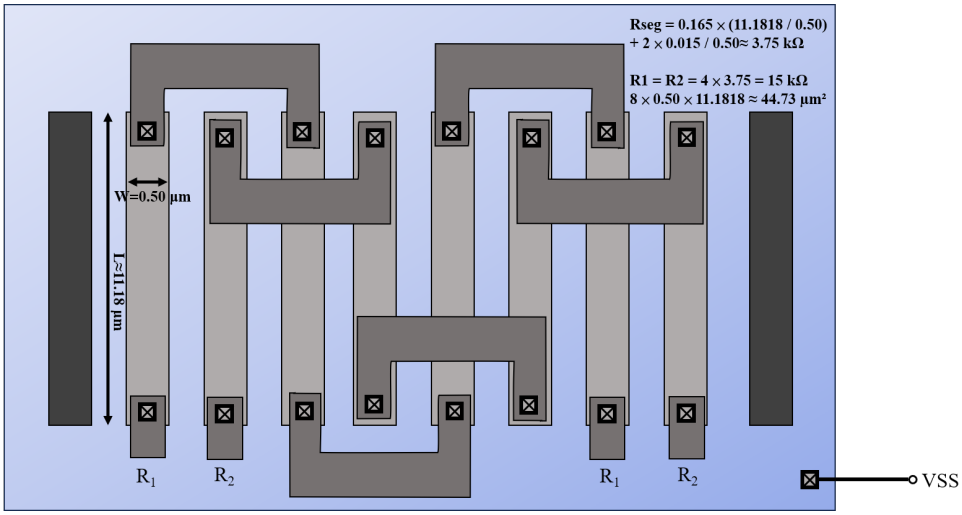


Figure 3: Interdigitized matched resistor pair with four series segments per resistor and one dummy strip at each outer edge.

(d) Matching and substrate connections. Keep segment geometry, contact counts, routing lengths, and the lengths of metal overlapping the resistor bodies balanced between the two groups. Where separate nets cross, use different metal layers without an interconnecting via, while balancing the layer usage and via counts between the groups.

Following Fig. 3-14 of the lab manual, leave each dummy's two signal terminals individually floating. All resistor backplates, including those of the dummies, connect to V_{SS}.

September 29, 2026

ECEN 474/704
Lab 5: Prelab
Anonymous contributor

1. **Comparison of current-mirror topologies.** Make a table listing the three current-mirror topologies described in this lab. Rate each topology using good, medium, and bad for output resistance, accuracy, complexity, and compliance voltage.

   Solution.

   The comparison assumes comparable device parameters and operating currents. "Good" denotes a desirable property: high output resistance, high accuracy, low complexity, or low required compliance voltage.

Topology	R_{out}	Accuracy	Complexity	Compliance voltage
Simple	Bad	Medium	Good	Good
Cascode	Good	Good	Medium	Bad
Low-voltage cascode	Good	Good	Bad	Medium

   (a) Simple current mirror. The two-transistor mirror has the lowest complexity and requires approximately one overdrive voltage at its output. Its output resistance is approximately r_o. Differences between the reference and output drain voltages produce current error through channel-length modulation.

   (b) Cascode current mirror. The cascode devices reduce changes in the drain voltages of the mirroring devices, improving current accuracy and increasing output resistance to approximately $g_m r_o^2$ when $g_m r_o \gg 1$. For equal overdrives, the conventional topology requires approximately $V_{TH} + 2V_{OV}$ of output voltage.

   (c) Low-voltage cascode current mirror. A separate bias branch sets the cascode gate voltage, allowing high output resistance and good current matching with an ideal minimum compliance voltage near $2V_{OV}$. This is lower than the conventional cascode requirement but higher than that of the simple mirror. The additional bias transistor and bias current increase circuit complexity. Saturation margin increases the practical compliance voltage above the ideal minimum.

2. **Simple 1:1 current-mirror design.** Design a simple 1:1 current mirror with a compliance voltage between 125 and 175 mV, an output current of 120 μA, and an output resistance of at least 25 kΩ. Determine the width and length of each transistor.

Solution.

(a) Process parameters and channel length. The single-finger $W/L = 5/0.27\,\mu$m NFET in the Lab 1 report provides the following extracted parameters:

$$\mu_n C_{ox} = 215.6\,\mu\text{A/V}^2, \quad V_{TH} = 0.433\,\text{V}, \quad \lambda = 0.1624\,\text{V}^{-1}.$$

For $L_1 = L_2 = 0.27\,\mu$m, the first-order output-resistance estimate is

$$R_{\text{out}} \approx \frac{1}{\lambda I_D} = \frac{1}{0.1624(120 \times 10^{-6})} = 51.3\,\text{k}\Omega > 25\,\text{k}\Omega.$$

(b) Overdrive and transistor widths. The center of the specified compliance range gives

$$V_{OV} \approx V_{\text{compliance}} = 0.150\,\text{V}, \qquad V_{GS} \approx V_{TH} + V_{OV} = 0.583\,\text{V}.$$

Neglecting channel-length modulation in the DC current, the square-law model gives

$$I_D \approx \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{OV}^2, \qquad \frac{W}{L} = \frac{2(120)}{215.6(0.150)^2} = 49.47.$$

Consequently,

$$\boxed{W_1 = W_2 = 13.36\,\mu\text{m}, \qquad L_1 = L_2 = 0.27\,\mu\text{m}.}$$

With $I_{\text{in}} = 120\,\mu$A, equal dimensions give a nominal 1:1 current ratio. The stated widths are total effective widths, not individual finger widths.

(c) Operating region and interpretation. M1 is diode-connected, with $V_{DS1} = V_{GS1} \approx 0.583\,\text{V} > V_{OV1}$. M2 is in saturation for approximately $V_{\text{out}} \geq 0.150\,\text{V}$ under this model.

The dimensions follow from a first-order model whose extracted parameters depend on geometry and bias. Mobility degradation, characterized by $\theta = 0.9999\,\text{V}^{-1}$ in Lab 1, is omitted from the sizing equation. Channel-length modulation introduces an output-voltage dependence into the drain current. The approximate resistance $1/(\lambda I_D)$ also neglects the finite $(1 + \lambda V_{DS})$ correction. These effects require simulation to determine the actual compliance voltage, output current, and output resistance.

3. **Low-voltage cascode current-mirror design.** Design a low-voltage cascode current mirror with a 1:2 input-to-output current ratio, a 60 μA input current, a low-frequency output resistance greater than 1 MΩ, and a compliance voltage below 500 mV. Determine all transistor widths and lengths and the bias current. Validate saturation at the DC operating point.

Solution.

(a) Topology and design assumptions. In the topology of Fig. 5-3 in the Lab 5 manual, M1 and M2 are the lower mirroring devices; M3 and M4 are their respective upper cascode devices. The drain of M3 connects to the gates of M1 and M2. The diode-connected bias device M_B supplies the gate voltage of M3 and M4. The sources of M1, M2, and M_B connect to V_{SS}; all NFET bulk terminals connect to V_{SS}.

The design uses the Lab 1 parameters of Question 2, with

$$L_1 = L_2 = L_3 = L_4 = L_B = 0.27\,\mu\text{m}, \qquad V_{OV1} = V_{OV2} = V_{OV3} = V_{OV4} = 0.150\,\text{V}.$$

The analysis neglects body effect and mobility degradation. The square law determines the device dimensions, and λ determines the small-signal output-resistance estimate.

(b) Reference and output branch dimensions. For M1 and M3, $I_D = 60\,\mu$A:

$$\left(\frac{W}{L}\right)_{1,3} = \frac{2(60)}{215.6(0.150)^2} = 24.74, \qquad W_1 = W_3 = 6.68\,\mu\text{m}.$$

For M2 and M4, $I_D = 120\,\mu$A:

$$\left(\frac{W}{L}\right)_{2,4} = 49.47, \qquad W_2 = W_4 = 13.36\,\mu\text{m}.$$

The doubled output-branch widths provide the nominal 1:2 current ratio.

(c) Bias branch and compliance voltage. Setting $V_{DS1} \approx V_{DS2} \approx 0.250\,\text{V}$ provides 0.100 V of nominal saturation margin for the lower devices. The required cascode gate voltage is

$$V_B = V_{S3} + V_{GS3} = 0.250 + (0.433 + 0.150) = 0.833\,\text{V}.$$

Thus $V_{OVB} = V_B - V_{TH} = 0.400\,\text{V}$. For $I_{\text{bias}} = 60\,\mu$A, the bias-device dimensions are

$$\left(\frac{W}{L}\right)_B = \frac{2(60)}{215.6(0.400)^2} = 3.479, \qquad \boxed{W_B \approx 0.94\,\mu\text{m}.}$$

The predicted compliance voltage is

$$\boxed{V_{\text{compliance}} \approx V_{S4} + V_{OV4} = 0.250 + 0.150 = 0.400\,\text{V} < 0.500\,\text{V}.}$$

(d) Saturation verification at the DC operating point. At the selected operating point, $V_{\text{out}} = 1.000\,\text{V}$. Ideal current sources supply the input and bias currents. Since the drain of M3 is connected to the gate of M1, $V_{D3} = V_{G1} \approx 0.583\,\text{V}$. The first-order saturation condition is $V_{DS} \geq V_{GS} - V_{TH} = V_{OV}$.

Device	V_{DS} (V)	V_{OV} (V)	Saturation check
M1	0.250	0.150	$0.250 > 0.150$
M2	0.250	0.150	$0.250 > 0.150$
M3	$0.583 - 0.250 = 0.333$	0.150	$0.333 > 0.150$
M4	$1.000 - 0.250 = 0.750$	0.150	$0.750 > 0.150$
M_B	0.833	0.400	$0.833 > 0.400$

All five devices satisfy the saturation condition under the stated first-order assumptions.

(e) Low-frequency output resistance. For the output devices M2 and M4,

$$r_{o2} \approx r_{o4} \approx \frac{1}{0.1624(120 \times 10^{-6})} = 51.3\,\text{k}\Omega, \qquad g_{m4} \approx \frac{2I_D}{V_{OV4}} = 1.60\,\text{mS}.$$

With the cascode gate held at its small-signal bias voltage and neglecting body effect,

$$\begin{aligned} R_{\text{out}} &\approx r_{o2} + r_{o4} + g_{m4} r_{o2} r_{o4} \\ &\approx \boxed{4.32\,\text{M}\Omega > 1\,\text{M}\Omega}. \end{aligned}$$

(f) Design summary and model validity.

Device	Total W (μm)	L (μm)	Nominal I_D (μA)
M1	6.68	0.27	60
M2	13.36	0.27	120
M3	6.68	0.27	60
M4	13.36	0.27	120
M_B	0.94	0.27	60

The nominal bias current is $I_{\text{bias}} = 60\,\mu\text{A}$. Because M3 and M4 have elevated source potentials with their bulks at V_{SS}, body effect raises their thresholds and can reduce the lower devices' saturation margin relative to this calculation. The Lab 1 summary does not provide body-effect parameters, so its effect on the operating point remains unquantified. Simulation of the device operating points determines whether W_B, I_{bias}, or other dimensions require adjustment. The tabulated values therefore represent analytical design estimates.

October 9, 2026

ECEN 474/704
Lab 6: Prelab
Anonymous contributor

1. **Comparison of differential-amplifier topologies.** Compare the two single-ended differential amplifiers in this lab. Rate differential gain, common-mode gain, output voltage swing, CMRR, and common-mode input voltage range. Include expressions for each specification.

Solution.

The comparison assumes comparable currents and device parameters. "Good" denotes larger differential gain, smaller common-mode gain magnitude, larger voltage range, or higher CMRR.

Specification	Simple (Fig. 6-6)	Cascode (Fig. 6-8)
Differential gain	Medium	Good
Common-mode gain ($\lvert A_{CM}\rvert$)	Medium	Good
Output voltage swing	Good	Bad
CMRR	Medium	Good
Common-mode input voltage range	Good	Bad

With $V_{DM} = V_{i+} - V_{i-}$ and $V_{CM} = (V_{i+} + V_{i-})/2$, zero differential input gives $I_{D1} = I_{D2} = I_{\text{tail}}/2$. The overdrive voltages are $V_{OVn} = V_{GS} - V_{TN}$ and $V_{OVp} = V_{SG} - |V_{TP}|$, both positive.

(a) Differential gain. The single-ended gain has no factor of $1/2$ because the PMOS mirror combines the two input-branch signal currents:

$$\begin{aligned} A_{DM,\text{simple}} &\approx g_{m1}(r_{o2} \parallel r_{o6}), \\ R_{P,\text{cas}} &\approx r_{o8} + r_{o10} + g_{m8} r_{o8} r_{o10}, \\ A_{DM,\text{cas}} &\approx g_{m1}(r_{o2} \parallel R_{P,\text{cas}}) \approx g_{m1} r_{o2}. \end{aligned}$$

The cascode increases the load resistance, but M2 is not cascoded; its r_{o2} therefore limits the differential gain.

(b) Common-mode gain. Using the low-frequency approximations in the Lab 6 handout,

$$\begin{aligned} A_{CM,\text{simple}} &\approx -\frac{1}{2 r_{o4} g_{m5}}, \\ R_{SS,\text{cas}} &\approx r_{o4} + r_{o6} + g_{m4} r_{o4} r_{o6}, \\ A_{CM,\text{cas}} &\approx -\frac{1}{2 R_{SS,\text{cas}} g_{m9}}. \end{aligned}$$

The cascode tail source has a higher resistance and reduces $|A_{CM}|$.

(c) Common-mode rejection ratio. For either topology,

$$\boxed{\text{CMRR} = 20 \log_{10} \left| \frac{A_{DM}}{A_{CM}} \right|.}$$

(d) Common-mode input voltage range. For the simple amplifier, the saturation constraints give

$$\begin{aligned} V_{CM,\min} &= V_{SS} + V_{OV4} + V_{TN1} + V_{OV1}, \\ V_{CM,\max} &= V_{DD} - |V_{TP5}| - V_{OV5} + V_{TN1}, \\ \Delta V_{CM} &= V_{CM,\max} - V_{CM,\min}. \end{aligned}$$

For the cascode amplifier, the voltage reserved across the lower NMOS mirror device is $d_n = V_{S4} - V_{SS}$. With a fixed V_{BN},

$$\begin{aligned} d_n &= V_{BN} - V_{SS} - V_{TN4} - V_{OV4}, \\ V_{CM,\min} &= V_{SS} + d_n + V_{OV4} + V_{TN1} + V_{OV1}, \\ V_{CM,\max} &= V_{DD} - |V_{TP9}| - V_{OV9} + V_{TN1}. \end{aligned}$$

The handout's minimum-headroom expression is the special case $d_n = V_{OV6}$. A practical choice $d_n > V_{OV6}$ provides saturation margin but raises the lower input limit. These ranges also require the reference branches and output devices to remain in saturation; output bias can impose an additional input limit.

(e) Output voltage swing. Let V_S be the common source voltage of M1 and M2. At a fixed common-mode input and with matched input devices,

$$V_{o,\min} = V_S + V_{OV2} = V_{CM} - V_{TN1}.$$

For the simple circuit, the upper output limit is $V_{o,\max} = V_{DD} - V_{OV6}$. For the cascode circuit, $d_p = V_{DD} - V_{S8}$ gives

$$d_p = V_{DD} - V_{BP} - |V_{TP8}| - V_{OV8}, \qquad V_{o,\max} = V_{DD} - d_p - V_{OV8}.$$

The ideal choice $d_p = V_{OV10}$ gives $V_{o,\max} = V_{DD} - V_{OV10} - V_{OV8}$. In both cases the available interval width is $\Delta V_o = V_{o,\max} - V_{o,\min}$, while the largest symmetric signal about $V_{o,Q}$ has peak amplitude

$$v_{o,\mathrm{pk}} < \min(V_{o,Q} - V_{o,\min},\, V_{o,\max} - V_{o,Q}).$$

These are saturation-based bounds; a specified distortion limit would require additional margin inside the interval.

Parameters and assumptions for Questions 2 and 3. The single-finger $W/L = 5/0.27\,\mu$m data in Table I of the Lab 1 report provide the device parameters:

Parameter	NFET	PFET
$K_P = \mu C_{ox}$ (μA/V^2)	215.6	47.57
$\lvert V_{TH}\rvert$ (V)	0.4330	0.4068
λ (V^{-1})	0.1624	0.1945
θ (V^{-1})	0.9999	0.6558

All initial lengths are $0.27\,\mu$m. The DC sizing uses the square-law model and neglects body effect, mobility degradation, and channel-length-modulation corrections to the drain current. The parameter λ enters only the small-signal estimate $r_o \approx 1/(\lambda I_D)$. All widths are total effective widths. The resulting dimensions constitute an analytical starting point.

2. **Simple differential-amplifier design (Fig. 6-6).** Design for SR $> 15\,\mathrm{V}/\mu\mathrm{s}$, GBW $> 8\,\mathrm{MHz}$, and $\Delta V_{CM} > 0.4\,\mathrm{V}$, with $V_{DD} = 0.9\,\mathrm{V}$, $V_{SS} = -0.9\,\mathrm{V}$, and $C_L = 10\,\mathrm{pF}$.

Solution.

(a) Tail current and input transconductance. The slew-rate constraint requires

$$I_{\mathrm{tail}} > C_L(15 \times 10^6) = 150\,\mu\mathrm{A}.$$

The handout gives angular bandwidth $\omega_{GBW} = g_{m1}/C_L$. Converting to hertz gives

$$\mathrm{GBW} = \frac{g_{m1}}{2\pi C_L}, \qquad g_{m1} > 2\pi(8 \times 10^6)(10 \times 10^{-12}) = 0.503\,\mathrm{mS}.$$

For $I_{\mathrm{tail}} = I_{\mathrm{bias}} = 180\,\mu\mathrm{A}$ and a 1:1 M3–M4 mirror, each input and PMOS load device carries $90\,\mu\mathrm{A}$ at the balanced operating point. The selected overdrives are

$$V_{OV1,2} = 0.250\,\mathrm{V}, \quad V_{OV3,4} = 0.150\,\mathrm{V}, \quad V_{OV5,6} = 0.200\,\mathrm{V}.$$

Then $g_{m1} = 2(90\,\mu\mathrm{A})/(0.250\,\mathrm{V}) = 0.720\,\mathrm{mS}$, providing margin above the required transconductance.

(b) Transistor widths and lengths. The square-law sizing relations are

$$I_D \approx \frac{1}{2}K_P\frac{W}{L}V_{OV}^2, \qquad \frac{W}{L} = \frac{2I_D}{K_P V_{OV}^2}, \qquad W = L\frac{W}{L}.$$

The sizing calculations, with currents in $\mu\mathrm{A}$, are

$$(W/L)_{1,2} = \frac{2(90)}{215.6(0.250)^2} = 13.358,$$
$$(W/L)_{3,4} = \frac{2(180)}{215.6(0.150)^2} = 74.212,$$
$$(W/L)_{5,6} = \frac{2(90)}{47.57(0.200)^2} = 94.597.$$

Device	I_D (μA)	V_{OV} (V)	W (μm)	L (μm)
M1, M2, each	90	0.250	3.607	0.270
M3, M4, each	180	0.150	20.037	0.270
M5, M6, each	90	0.200	25.541	0.270

NMOS bodies connect to V_{SS} and PMOS bodies to V_{DD}. Matched device pairs use equal dimensions and consistent finger configurations. The diode-connected M3 establishes the common gate voltage, eliminating the need for a separate tail-bias voltage source.

(c) Common-mode range. Substituting the selected overdrives gives

$$V_{CM,\mathrm{min}} = -0.9 + 0.150 + 0.433 + 0.250 = -0.067\,\mathrm{V},$$
$$V_{CM,\mathrm{max}} = 0.9 - 0.4068 - 0.200 + 0.433 = 0.7262\,\mathrm{V},$$
$$\Delta V_{CM} = \boxed{0.7932\,\mathrm{V} > 0.4\,\mathrm{V}.}$$

The selected bias $V_{CM,Q} = 0.300\,\mathrm{V}$ lies inside this interval.

(d) Nominal DC operating point and saturation checks. For equal inputs, a symmetry-consistent operating point places the output at the diode-connected PMOS voltage:

$$\begin{aligned} V_S &= 0.300 - (0.433 + 0.250) = -0.3830\,\text{V}, \\ V_{D1} = V_{o,Q} &= 0.900 - (0.4068 + 0.200) = 0.2932\,\text{V}, \\ V_{G3} = V_{G4} &= -0.900 + 0.433 + 0.150 = -0.3170\,\text{V}. \end{aligned}$$

Each device satisfies $V_{DS} > V_{OV}$ for an NMOS or $V_{SD} > V_{OV}$ for a PMOS under the assumed model:

Device	V_{DS} or V_{SD} (V)	V_{OV} (V)	Saturation
M1, M2	0.6762	0.250	Yes
M3	0.5830	0.150	Yes
M4	0.5170	0.150	Yes
M5, M6	0.6068	0.200	Yes

At $V_{CM,Q} = 0.300\,\text{V}$, the output limits are

$$-0.133\,\text{V} < V_o < 0.700\,\text{V}.$$

The interval is 0.833 V wide. About $V_{o,Q} = 0.2932\,\text{V}$, the symmetric-swing limit approaches 0.4068 V peak. A finite saturation margin reduces the usable amplitude below this limit.

(e) Small-signal performance. Using the stated λ values,

$$\begin{aligned} r_{o2} &= \frac{1}{0.1624(90\times10^{-6})} = 68.418\,\text{k}\Omega, \\ r_{o6} &= \frac{1}{0.1945(90\times10^{-6})} = 57.127\,\text{k}\Omega, \\ r_{o4} &= \frac{1}{0.1624(180\times10^{-6})} = 34.209\,\text{k}\Omega, \\ R_{\text{out}} &= r_{o2} \parallel r_{o6} = 31.132\,\text{k}\Omega. \end{aligned}$$

Thus $A_{DM} = g_{m1}R_{\text{out}} = 22.415\,\text{V/V}$ (27.01 dB). With $g_{m5} = 2(90\,\mu\text{A})/(0.200\,\text{V}) = 0.900\,\text{mS}$,

$$A_{CM} \approx -\frac{1}{2r_{o4}g_{m5}} = -0.01624, \qquad \text{CMRR} \approx 62.80\,\text{dB}.$$

The dominant-pole approximation gives

$$f_{p1} = \frac{1}{2\pi R_{\text{out}}C_L} = 511.22\,\text{kHz}, \quad \text{GBW} = \frac{g_{m1}}{2\pi C_L} = 11.46\,\text{MHz}.$$

Specification	Requirement	Analytical estimate
Slew rate, I_{tail}/C_L	$> 15\,\text{V}/\mu\text{s}$	$18\,\text{V}/\mu\text{s}$
GBW	$> 8\,\text{MHz}$	11.46 MHz
Common-mode range width	$> 0.4\,\text{V}$	0.7932 V

3. **Differential amplifier with cascode mirrors (Fig. 6-8).** Design for SR $> 7\,\mathrm{V}/\mu\mathrm{s}$, GBW $> 8\,\mathrm{MHz}$, and CMRR > 65 dB, with $V_{DD} = 0.9\,\mathrm{V}$, $V_{SS} = -0.9\,\mathrm{V}$, and $C_L = 10\,\mathrm{pF}$. Determine V_{BP} and V_{BN} and retain a nonzero input common-mode range.

Solution.

(a) Current, overdrives, and transistor dimensions. The minimum tail current is $C_L(7 \times 10^6) = 70\,\mu\mathrm{A}$; the input-transconductance minimum remains 0.503 mS. A 1:1 tail mirror establishes

$$I_{\mathrm{bias}} = I_{\mathrm{tail}} = 120\,\mu\mathrm{A}, \qquad I_{D1,2,7,8,9,10} = 60\,\mu\mathrm{A}.$$

For $V_{OV} = 0.150\,\mathrm{V}$ in all ten devices and $L = 0.270\,\mu\mathrm{m}$, the sizing equation of Question 2 gives

$$(W/L)_{1,2} = \frac{2(60)}{215.6(0.150)^2} = 24.737,$$
$$(W/L)_{3,4,5,6} = \frac{2(120)}{215.6(0.150)^2} = 49.474,$$
$$(W/L)_{7,8,9,10} = \frac{2(60)}{47.57(0.150)^2} = 112.115.$$

Device	I_D (μA)	V_{OV} (V)	W (μm)	L (μm)
M1, M2, each	60	0.150	6.679	0.270
M3–M6, each	120	0.150	13.358	0.270
M7–M10, each	60	0.150	30.271	0.270

The input transconductance is $g_{m1} = 2(60\,\mu\mathrm{A})/(0.150\,\mathrm{V}) = 0.800\,\mathrm{mS}$.

(b) Cascode bias voltages. Reserving $d_n = d_p = 0.200\,\mathrm{V}$ across the lower NMOS and upper PMOS mirror devices gives 50 mV of nominal saturation margin above their 0.150 V overdrives:

$$V_N = V_{S3} = V_{S4} = V_{SS} + d_n = -0.700\,\mathrm{V},$$
$$V_P = V_{S7} = V_{S8} = V_{DD} - d_p = 0.700\,\mathrm{V},$$
$$V_{BN} = V_N + V_{TN4} + V_{OV4} = \boxed{-0.117\,\mathrm{V},}$$
$$V_{BP} = V_P - |V_{TP8}| - V_{OV8} = \boxed{0.1432\,\mathrm{V}.}$$

As permitted by the handout, external sources at the testbench supply these biases through additional schematic pins. The amplifier therefore requires no internal bias-generation transistors or additional bias current for these voltages.

(c) Common-mode input range with these biases. Including the selected d_n, rather than only its ideal minimum,

$$V_{CM,\mathrm{min}} = -0.900 + 0.200 + 0.150 + 0.433 + 0.150 = 0.0330\,\mathrm{V},$$
$$V_{CM,\mathrm{max}} = 0.900 - 0.4068 - 0.150 + 0.433 = 0.7762\,\mathrm{V}.$$

Hence $\boxed{\Delta V_{CM} = 0.7432\,\mathrm{V} > 0}$. The nominal operating point is $V_{CM,Q} = 0.300\,\mathrm{V}$.

(d) DC operating point and saturation checks. In Fig. 6-8, M3's drain drives the gates of M5 and M6; M7's drain drives the gates of M9 and M10. Therefore,

$$V_{G5} = V_{G6} = V_{D3} = -0.900 + 0.433 + 0.150 = -0.3170\,\text{V},$$
$$V_{D1} = V_{G9} = V_{G10} = V_{o,Q} = 0.900 - 0.4068 - 0.150 = 0.3432\,\text{V},$$
$$V_S = 0.300 - 0.433 - 0.150 = -0.2830\,\text{V}.$$

Together with $V_N = -0.700\,\text{V}$ and $V_P = 0.700\,\text{V}$, these voltages give the following first-order saturation checks:

Device	V_{DS} or V_{SD} (V)	V_{OV} (V)	Saturation
M1, M2	0.6262	0.150	Yes
M3	0.3830	0.150	Yes
M4	0.4170	0.150	Yes
M5, M6	0.2000	0.150	Yes
M7, M8	0.3568	0.150	Yes
M9, M10	0.2000	0.150	Yes

At the chosen common-mode voltage, the output interval is $-0.133\,\text{V} < V_o < 0.550\,\text{V}$. Its width is 0.683 V, and the symmetric swing about $V_{o,Q} = 0.3432\,\text{V}$ approaches 0.2068 V peak. The smaller upper headroom is the cost of the cascode PMOS load.

(e) Gain, common-mode rejection, and frequency estimates. The input and PMOS transconductances are 0.800 mS; the NMOS tail-device transconductances are 1.600 mS. Using $r_o \approx 1/(\lambda I_D)$ gives

$$r_{o2} = 102.627\,\text{k}\Omega, \quad r_{o4} = r_{o6} = 51.314\,\text{k}\Omega, \quad r_{o8} = r_{o10} = 85.690\,\text{k}\Omega.$$

Consequently,

$$R_{SS} \approx r_{o4} + r_{o6} + g_{m4} r_{o4} r_{o6} = 4.3156\,\text{M}\Omega,$$
$$R_P \approx r_{o8} + r_{o10} + g_{m8} r_{o8} r_{o10} = 6.0456\,\text{M}\Omega,$$
$$R_{\text{out}} \approx r_{o2} \parallel R_P = 100.914\,\text{k}\Omega,$$
$$A_{DM} \approx g_{m1} R_{\text{out}} = 80.731\ \text{V/V} \quad (38.14\,\text{dB}),$$
$$A_{CM} \approx -\frac{1}{2 R_{SS} g_{m9}} = -1.4482 \times 10^{-4},$$
$$\text{CMRR} \approx 20 \log_{10} \left| \frac{80.731}{-1.4482 \times 10^{-4}} \right| = 114.92\,\text{dB}.$$

Specification	Requirement	Analytical estimate
Slew rate, I_{tail}/C_L	$> 7\,\text{V}/\mu\text{s}$	$12\,\text{V}/\mu\text{s}$
GBW, $g_{m1}/(2\pi C_L)$	$> 8\,\text{MHz}$	12.73 MHz
CMRR	> 65 dB	114.92 dB
Common-mode range width	Nonzero	0.7432 V
Dominant pole, $1/(2\pi R_{\text{out}} C_L)$	Record estimate	157.71 kHz

Model validity and comparison with simulation results.
The calculated dimensions satisfy the requirements within the stated first-order model. The Lab 1 parameters were extracted at one geometry and bias condition. Body effect, mobility degradation, finite drain-voltage-dependent currents, internal poles, and parasitic capacitances are not fully represented by these estimates. Pole and GBW calculations assume the 10 pF load dominates the output capacitance and higher-frequency internal poles are negligible. The Lab 1 report provides θ but no numerical body-effect parameters. Threshold shifts caused by the elevated NMOS source voltages and PMOS cascode source voltages therefore remain unquantified. The computed CMRR assumes matched devices and ideal AC-grounded cascode biases, excluding mismatch effects.
The Lab 5 report illustrates the difference between analytical sizing and simulation-refined dimensions. Its simulated $120\,\mu$A simple mirror uses $W = 19.4\,\mu$m, $L = 0.270\,\mu$m for approximately 0.150 V compliance, whereas the first-order square-law calculation gives $W = 13.36\,\mu$m. The final cascode mirror uses $L = 0.320\,\mu$m to increase output resistance. These dimensions characterize the refined Lab 5 circuits; they do not establish a new device-parameter set for the Lab 6 designs. The present design retains $L = 0.270\,\mu$m to use the corresponding Lab 1 parameter set without assigning an unmeasured λ to $0.320\,\mu$m devices.
A common-mode DC sweep determines the interval in which all active devices remain in saturation. The widths, reference current, and V_{BN}/V_{BP} control the bias point and provide the variables for design refinement. Separate differential-mode and common-mode AC excitations determine the corresponding gains, CMRR, GBW, and dominant pole. Slew rate requires a large-signal transient analysis because it lies outside the linearized AC model. Saturation and performance margins account for the effects omitted from the first-order design.

References.

[1] S. Palermo, *ECEN 474/704 Lab 6: Differential Pairs*, course laboratory handout, Figs. 6-6 and 6-8 and Prelab Questions 1–3.

[2] Anonymous contributor, *ECEN 704 Lab 1 Report*, Table I: single-finger $5/0.27\,\mu$m NFET and PFET parameter rows.

[3] Anonymous contributor, *ECEN 704 Lab 5 Report*, Sec. III and Table 3: final mirror sizes and simulation-based refinement.

October 16, 2026

ECEN 474/704
Lab 7: Prelab
Anonymous contributor

1. **Comparison of inverting-amplifier topologies.** Rank the four amplifiers in gain, input impedance, output impedance, and linearity. Give expressions for the first three quantities.

   Solution.

   The comparison assumes comparable bias currents and device parameters. The ratings favor high voltage gain, high input impedance, and low output impedance for driving a load.

Specification	Current mirror	Digital CMOS	PMOS buffer	CMOS buffer
Voltage gain	Medium	Good	Poor	Poor
Input impedance	Good	Good	Good	Good
Output impedance	Poor	Poor	Good	Good
Linearity	Medium	Poor	Good	Medium

   The buffers have a target gain of -1 and are linear in the ideal square-law model. Tracking between devices of the same type favors the PMOS pair, although body effect and other non-idealities limit its linearity.

   (a) Gain and output resistance. Let $g_{oi} = 1/r_{oi}$ and $g_o = g_{o1} + g_{o2}$. With the supplies and bias-reference node at AC ground,

Circuit	A_{v0}	R_{out}
Current-mirror load	$-g_{m1}/g_o$	$1/g_o$
Digital CMOS	$-(g_{m1} + g_{m2})/g_o$	$1/g_o$
Self-biased PMOS	$-g_{m1}/(g_{m2} + g_{mb2} + g_o)$	$1/(g_{m2} + g_{mb2} + g_o)$
Self-biased CMOS	$-g_{m1}/(g_{m2} + g_o)$	$1/(g_{m2} + g_o)$

   The output resistance is evaluated with the input held constant. A finite R_L adds $1/R_L$ to each denominator. The PMOS load's source is the output while its body is fixed, giving the g_{mb2} term. The CMOS load's source and body are both fixed at V_{DD}.

   (b) Input impedance. Ignoring gate leakage, all four circuits have $R_{\text{in,DC}} \to \infty$. At finite frequency, let $k(s) = v_o/v_A$, where v_A is the gate node after R_S. Then

$$Z_{\text{in},A}(s) = \frac{1}{s\{C_A + C_C[1 - k(s)]\}} \approx \frac{1}{s[C_A + C_C(1 + |A_{v0}|)]}.$$

   The source sees $R_S + Z_{\text{in},A}$. For the single-input-device circuits, $C_A \approx C_{gs1}$; digital CMOS has $C_A \approx C_{gs1} + C_{gs2}$. Similarly, $C_C = C_m + C_{gd1}$, with C_{gd2} added for digital CMOS. Thus the equal DC input-resistance ratings do not imply equal AC input loading.

2. **Miller-theorem derivation and comparison.** Derive the generic amplifier transfer function using Miller's theorem, compare it with the nodal result, and discuss the approximation's utility.

Solution.

(a) Miller equivalent. Let v_A be the input node after R_A, let v_o be the output, and define $A_0 = G_m R_B > 0$, so the DC voltage gain is $-A_0$. For the capacitor C_C between these nodes,

$$i_A = sC_C(v_A - v_o) = sC_C[1 - k(s)]v_A,$$
$$i_o = sC_C(v_o - v_A) = sC_C[1 - 1/k(s)]v_o, \qquad k(s) = \frac{v_o}{v_A}.$$

These identities are exact if the frequency-dependent $k(s)$ is retained. Replacing $k(s)$ by its low-frequency value $-A_0$ gives

$$\boxed{C_{\text{in}} = C_A + C_C(1 + A_0), \qquad C_{\text{out}} = C_B + C_C(1 + 1/A_0).}$$

The input divider and output transconductance stage then give

$$\frac{v_A}{v_{\text{in}}} \approx \frac{1}{1 + sR_A C_{\text{in}}}, \qquad \frac{v_o}{v_A} \approx \frac{-G_m R_B}{1 + sR_B C_{\text{out}}}.$$

Thus the constant-gain Miller approximation is

$$\boxed{H_M(s) \approx \frac{-A_0}{(1 + sR_A C_{\text{in}})(1 + sR_B C_{\text{out}})}.}$$

The resulting time constants are $R_A C_{\text{in}}$ and $R_B C_{\text{out}}$. Their accuracy as pole estimates depends on the frequency variation of the inter-node gain.

(b) Exact nodal result. With the controlled current $G_m v_A$ directed from the output to AC ground, the two node equations are

$$\left[\frac{1}{R_A} + s(C_A + C_C)\right] v_A - sC_C v_o = \frac{v_{\text{in}}}{R_A},$$
$$(G_m - sC_C)v_A + \left[\frac{1}{R_B} + s(C_B + C_C)\right] v_o = 0.$$

Eliminating v_A yields

$$\boxed{H(s) = \frac{-A_0(1 - sC_C/G_m)}{1 + as + bs^2},}$$

where

$$a = R_A(C_A + C_C) + R_B(C_B + C_C) + G_m R_A R_B C_C,$$
$$b = R_A R_B(C_A C_B + C_A C_C + C_B C_C).$$

The denominator and zero agree with the handout. The overall minus sign makes $H(0) = -G_m R_B$, consistent with an inverting amplifier. The exact inter-node gain is

$$k(s) = -\frac{G_m - sC_C}{1/R_B + s(C_B + C_C)}.$$

Using this $k(s)$ in the Miller identities recovers the same nodal result.

(c) Comparison and utility. Expanding H_M gives coefficients $a_M = R_A C_{\text{in}} + R_B C_{\text{out}}$ and $b_M = R_A R_B C_{\text{in}} C_{\text{out}}$. Their differences from the exact denominator are

$$a_M - a = \frac{R_B C_C}{A_0} = \frac{C_C}{G_m},$$
$$b_M - b = R_A R_B \left[\frac{C_A C_C}{A_0} + A_0 C_B C_C + C_C^2 \left(2 + A_0 + \frac{1}{A_0} \right) \right].$$

The constant-gain approximation also removes the right-half-plane zero

$$\boxed{s_z = +\frac{G_m}{C_C}, \qquad f_z = \frac{G_m}{2\pi C_C}.}$$

Miller's theorem itself is an exact circuit identity; the loss of accuracy comes from substituting the constant $-A_0$ for $k(s)$. The approximation therefore provides an estimate of input loading and an initial compensation capacitance. The second pole and phase near unity gain require the frequency-dependent gain or the nodal result.

For $A_0 \gg 1$, negligible C_A, and a dominant Miller time constant,

$$f_{p1} \approx \frac{1}{2\pi A_0 R_A C_C}, \qquad \text{GBW} \approx A_0 f_{p1} \approx \frac{1}{2\pi R_A C_C}.$$

For $R_A = 100\,\text{k}\Omega$ and GBW $= 2\,\text{MHz}$, this gives

$$C_C \approx 0.7958\,\text{pF}.$$

This is the total feedback capacitance, including C_{gd}; the external C_m is smaller by the relevant parasitic capacitance. The full two-node expression provides the corresponding refinement.

(d) Frequency and phase conventions. When $a^2 \geq 4b$, the two real poles are at $s = -\omega_{p1}$ and $s = -\omega_{p2}$, where

$$\omega_{p1} = \frac{2}{a + \sqrt{a^2 - 4b}}, \qquad \omega_{p2} = \frac{a + \sqrt{a^2 - 4b}}{2b}, \qquad f_{pi} = \frac{\omega_{pi}}{2\pi}.$$

Define the measured bandwidth f_{BW} by $|H(j2\pi f_{\text{BW}})| = A_0/\sqrt{2}$. It is approximately f_{p1} only when the other pole and the zero are sufficiently far away. The measured gain–bandwidth product is GBW $= A_0 f_{\text{BW}}$, while the unity-gain frequency satisfies $|H(j2\pi f_{\text{UGF}})| = 1$.

After removing the DC inversion, the dynamic phase lag is

$$\phi_{\text{lag}}(f) = \tan^{-1}\left(\frac{f}{f_{p1}}\right) + \tan^{-1}\left(\frac{f}{f_{p2}}\right) + \tan^{-1}\left(\frac{f}{f_z}\right).$$

The RHP zero adds phase lag. Therefore, $f_{p2} \geq \sqrt{3}\,\text{GBW}$ alone guarantees neither the exact UGF nor a lag below 120° when the zero is appreciable. For a monotone low-pass buffer with $A_0 = 1$, a positive-frequency unity-gain crossing need not exist. The buffer phase is therefore evaluated at f_{BW}, as specified in the handout.

3. **Design of the four inverting amplifiers.** Design the current-mirror-loaded inverter, digital CMOS inverter, self-biased PMOS inverter, and self-biased CMOS inverter in Figs. 7-5, 7-6, 7-8, and 7-9, respectively. All four use $V_{DD} = +0.9\,\text{V}$, $V_{SS} = -0.9\,\text{V}$, $R_S = 100\,\text{k}\Omega$, and GBW = 2 MHz.

Solution.

Common specifications and device assumptions. The high-gain designs require $A_0 = 10^{30/20} = 31.6228$, $C_L = 1\,\text{pF}$, and dynamic phase lag below 120° at UGF. The two diode-load designs require $A_0 = 1$ and $C_L = 25\,\text{pF}$. Here $A_0 = |A_{v0}|$ is positive; the voltage gain is negative. The nominal output is selected as $V_{o,Q} = 0$.

The device parameters are taken from Table I of the Lab 1 MOS characterization report, measured at single-finger $W/L = 5/0.27\,\mu\text{m}$:

Parameter	NFET	PFET
$K_P = \mu C_{ox}$ ($\mu\text{A/V}^2$)	215.6	47.57
$\lvert V_{TH}\rvert$ (V)	0.4330	0.4068
λ at $L_0 = 0.27\,\mu\text{m}$ (V^{-1})	0.1624	0.1945
θ (V^{-1})	0.9999	0.6558

The DC sizing uses the square-law model. Channel-length modulation is retained only through the small-signal estimate $g_o \approx \lambda I_D$:

$$I_D = \frac{1}{2} K_P \frac{W_{\text{eff}}}{L} V_{OV}^2, \qquad g_m = \frac{2I_D}{V_{OV}}, \qquad r_o \approx \frac{1}{\lambda I_D}.$$

The numerical designs initially omit body effect, mobility degradation, and DC channel-length-modulation corrections. The listed θ values are not applied to the square-law results. For lengths different from L_0, the assumed scaling is

$$\lambda_n(L) = 0.1624 \frac{L_0}{L}, \qquad \lambda_p(L) = 0.1945 \frac{L_0}{L},$$

while holding K_P and V_{TH} fixed. This length dependence is a design approximation that has not been verified by device characterization.

The frequency calculation assumes $C_A = 0$, $C_B = C_L$, and $C_C = C_m$. Gate, junction, and interconnect capacitances in the PDK modify this model. The results below are analytical estimates; their accuracy requires circuit simulation.

Width and multiplicity convention. Each table gives both total effective width and width per finger:

$$W_{\text{eff}} = w\, n_f\, m.$$

Parts (a)–(d) derive continuous w_{ana} and L_{ana}. The later grid-constrained table gives the initial dimensions, with w, L on the $0.01\,\mu\text{m}$ grid and initial $n_f = m = 2$; both counts remain adjustable. DC and AC verification apply to the selected geometry; any subsequent dimensional change requires recalculation and simulation.

(a) Current-mirror-loaded inverter (Fig. 7-5).

Transconductance and bias current. M1 is the NMOS driver; M2 and M3 form a 1:1 PMOS mirror. The handout's second-pole estimate would give

$$g_{m1} \gtrsim 2\pi(\sqrt{3})(2\,\text{MHz})(1\,\text{pF}) = 21.77\,\mu\text{S}.$$

This bound neglects the RHP zero. A larger value, $g_{m1} = 300\,\mu\text{S}$, is selected, with phase evaluated using the complete two-node transfer function. At $L = 0.270\,\mu\text{m}$, $\lambda_\Sigma = \lambda_n + \lambda_p = 0.3569\,\text{V}^{-1}$. The required current and overdrive are

$$I_D = I_{\text{bias}} = \frac{g_{m1}}{A_0\lambda_\Sigma} = \frac{300\times10^{-6}}{31.6228(0.3569)} = 26.581\,\mu\text{A},$$
$$R_{\text{out}} = \frac{1}{\lambda_\Sigma I_D} = 105.409\,\text{k}\Omega,$$
$$V_{OV1} = \frac{2I_D}{g_{m1}} = 0.177208\,\text{V}.$$

The choice $g_{m2} = g_{m3} = g_{m1}$ gives equal overdrives for all three devices. The required aspect ratios are

$$\left(\frac{W_{\text{eff}}}{L}\right)_n = \frac{2(26.581)}{215.6(0.177208)^2} = 7.8522, \quad \left(\frac{W_{\text{eff}}}{L}\right)_p = 35.5881.$$

Device	w_{ana} (μm)	L_{ana} (μm)	n_f	m	W_{eff} (μm)
M1 (NFET)	0.53002	0.27000	2	2	2.12008
M2, M3 (PFET), each	2.40220	0.27000	2	2	9.60878

The estimated gate biases are

$$V_{in,Q} = V_{SS} + V_{TN} + V_{OV1} = -0.289792\,\text{V},$$
$$V_{G2} = V_{G3} = V_{DD} - |V_{TP}| - V_{OV2} = 0.315992\,\text{V}.$$

With $V_{o,Q} = 0$, M1 and M2 each have $|V_{DS}| = 0.9\,\text{V}$; M3 is diode-connected with $V_{SD3} = 0.584008\,\text{V}$. All exceed $V_{OV} = 0.177208\,\text{V}$ under the assumed model. The mirror current will require adjustment when unequal drain voltages and DC channel-length modulation are included.

Compensation and two-node response. The Miller estimate gives $C_C \approx 0.795775\,\text{pF}$. With $f_{\text{BW}} = 2\,\text{MHz}/31.6228 = 63.2456\,\text{kHz}$, the complete transfer function satisfies the half-power condition at $\omega_B = 2\pi f_{\text{BW}}$:

$$(1 - b\omega_B^2)^2 + (a\omega_B)^2 = 2\left[1 + \left(\frac{\omega_B C_C}{G_m}\right)^2\right]. \tag{1}$$

This yields $C_m = C_C = 0.716828\,\text{pF}$ in the no-parasitic model. The resulting estimates are

$$A_{v0} = -31.6228, \quad f_{p1} = 63.2456\,\text{kHz}, \quad f_{p2} = 53.005\,\text{MHz}, \quad f_z = 66.608\,\text{MHz}.$$

The actual unity-gain crossing of this model is $f_u = 1.99848\,\text{MHz}$, where the dynamic phase lag is $92.07^\circ < 120^\circ$. Its GBW is 2.000 MHz. The external C_m is reduced by the intrinsic contribution of C_{gd1}.

(b) Digital CMOS inverter (Fig. 7-6).

Length and current selection. Both gates are driven by the input, giving $G_m = g_{mn} + g_{mp}$. The selected bias is $V_{in,Q} = V_{o,Q} = 0$, with equal lengths $L_n = L_p = L$. The overdrives are

$$V_{OVn} = 0.9 - 0.4330 = 0.4670\,\mathrm{V}, \qquad V_{OVp} = 0.9 - 0.4068 = 0.4932\,\mathrm{V}.$$

Equal drain currents require

$$\frac{\beta_p}{\beta_n} = \left(\frac{V_{OVn}}{V_{OVp}}\right)^2 = 0.89658, \qquad \beta_i = K_{Pi}\frac{W_{\mathrm{eff},i}}{L_i}.$$

The unequal thresholds mean that $\beta_n = \beta_p$ does not place the exact transition center at zero. Using the length-scaling assumption,

$$A_0 = \frac{2/V_{OVn} + 2/V_{OVp}}{\lambda_n(L) + \lambda_p(L)},$$
$$L = L_0\,\frac{A_0[\lambda_n(L_0) + \lambda_p(L_0)]}{2/V_{OVn} + 2/V_{OVp}} = 0.365476\,\mu\mathrm{m}.$$

The choice $I_D = 120\,\mu\mathrm{A}$ places the non-dominant pole and zero well above the required bandwidth in the assumed model. Then

$$g_{mn} = \frac{2(120\,\mu\mathrm{A})}{0.4670\,\mathrm{V}} = 513.919\,\mu\mathrm{S},$$
$$g_{mp} = \frac{2(120\,\mu\mathrm{A})}{0.4932\,\mathrm{V}} = 486.618\,\mu\mathrm{S},$$
$$G_m = 1000.537\,\mu\mathrm{S}, \qquad R_{\mathrm{out}} = \frac{A_0}{G_m} = 31.6058\,\mathrm{k\Omega}.$$

Device sizes and operating point. The effective width of each transistor follows from $W_{\mathrm{eff}} = 2I_D L/(K_P V_{OV}^2)$:

Device	w_{ana} (μm)	L_{ana} (μm)	n_f	m	W_{eff} (μm)
M1 (NFET)	0.46637	0.36548	2	2	1.86547
M2 (PFET)	1.89509	0.36548	2	2	7.58037

At the selected bias, $V_{DS1} = V_{SD2} = 0.9\,\mathrm{V}$, which exceeds both overdrives. The NMOS body is at V_{SS} and the PMOS body at V_{DD}; both active devices have zero source-body bias.

Compensation and performance. Solving Eq. (1) with the new G_m and R_{out} gives $C_m = C_C = 0.754672\,\mathrm{pF}$. The resulting estimates are

$$A_{v0} = -31.6228 \quad (30.000\,\mathrm{dB}),$$
$$f_{\mathrm{BW}} = 63.2456\,\mathrm{kHz}, \qquad \mathrm{GBW} = 2.000\,\mathrm{MHz},$$
$$f_{p2} = 167.913\,\mathrm{MHz}, \qquad f_z = 211.006\,\mathrm{MHz},$$
$$f_u = 1.99895\,\mathrm{MHz}, \qquad \phi_{\mathrm{lag}}(f_u) = 89.41^\circ.$$

The analytical response satisfies the phase requirement. In the physical circuit, $C_C = C_m + C_{gd1} + C_{gd2}$; gate and output capacitances also affect the AC response.

(c) PMOS-only inverter with self-biased load (Fig. 7-8).

Unity-gain sizing with finite output conductances. M1 is the upper driver. M2 has drain and gate at V_{SS}, source at V_o, and body at V_{DD}. For the initial no-body-effect calculation at $V_{o,Q} = 0$,

$$V_{OV2} = 0.9 - |V_{TP}| = 0.4932\,\mathrm{V}.$$

The handout's ideal equal-size result $V_o = -V_{in}$ neglects finite g_o. Retaining g_o gives

$$A_{v0} = -\frac{g_{m1}}{g_{m2} + g_{o1} + g_{o2}}.$$

Unity gain therefore requires $G_m = g_{m1} = G_{\mathrm{out}}$, giving $R_{\mathrm{out}} = 1/G_m$ and $A_{v0} = -1$. The simple load-dominant estimate is

$$G_m \approx 2\pi C_L \mathrm{GBW} = 314.159\,\mu\mathrm{S}.$$

For the initial choice $C_m = 0.010\,\mathrm{pF}$, Eq. (1) determines G_m at $f_{\mathrm{BW}} = 2\,\mathrm{MHz}$. For this unity-gain case,

$$a = 2R_S C_m + \frac{C_L + C_m}{G_m}, \qquad b = \frac{R_S C_L C_m}{G_m},$$

which gives $G_m = 318.387\,\mu\mathrm{S}$. The realizability of this small C_m depends on the PDK's available capacitance range and the intrinsic parasitic capacitance.

The choice $L_1 = L_2 = 0.270\,\mu\mathrm{m}$ gives $\lambda_\Sigma = 2\lambda_p = 0.3890\,\mathrm{V}^{-1}$. Since $g_{m2} = 2I_D/V_{OV2}$,

$$\begin{aligned} I_D &= \frac{G_m}{2/V_{OV2} + \lambda_\Sigma} = 71.642\,\mu\mathrm{A}, \\ V_{OV1} &= \frac{2I_D}{G_m} = 0.450030\,\mathrm{V}, \\ V_{in,Q} &= V_{DD} - |V_{TP}| - V_{OV1} = 0.043170\,\mathrm{V}. \end{aligned}$$

Applying the square-law sizing equation separately to M1 and M2:

Device	w_{ana} (μm)	L_{ana} (μm)	n_f	m	W_{eff} (μm)
M1 (PFET driver)	1.00389	0.27000	2	2	4.01555
M2 (PFET load)	0.83584	0.27000	2	2	3.34335

The driver is slightly wider to compensate the two output conductances. Here $g_{m2} = 290.519\,\mu\mathrm{S}$ and $g_{o1} + g_{o2} = 27.869\,\mu\mathrm{S}$. Both PMOS devices have $V_{SD} = 0.9\,\mathrm{V}$, exceeding their respective overdrives under the no-body-effect assumption.

The complete two-node calculation gives

$$A_{v0} = -1, \quad R_{\mathrm{out}} = 3.14083\,\mathrm{k\Omega}, \quad f_{\mathrm{BW}} = \mathrm{GBW} = 2.000\,\mathrm{MHz}.$$

Its pole frequencies are 2.00031 MHz and 161.272 MHz; the RHP zero is at 5067.29 MHz. The phase lag at f_{BW} is 45.73°. This unity-DC-gain low-pass model has no positive-frequency unity-gain crossing, so its phase is evaluated at the bandwidth.

Body-effect correction for the PMOS load. The preceding PMOS-only dimensions neglect body effect. With M2's body fixed at V_{DD} and its source at zero, $V_{BS2} = 0.9\,\mathrm{V}$. The load therefore exhibits body effect, whereas the upper driver has zero source-body bias. Let $\Phi = 2|\phi_F|$ and use the first-order relations

$$|V_{TP2}| \approx |V_{TP0}| + \gamma_p \left(\sqrt{\Phi + V_{BS2}} - \sqrt{\Phi}\right),$$
$$\eta = \frac{g_{mb2}}{g_{m2}} \approx \frac{\gamma_p}{2\sqrt{\Phi + V_{BS2}}}.$$

The corrected quantities become

$$V_{OV2} = 0.9 - |V_{TP2}|,$$
$$G_{\mathrm{out}} = g_{m2}(1+\eta) + \lambda_\Sigma I_D,$$
$$I_D = \frac{G_m}{2(1+\eta)/V_{OV2} + \lambda_\Sigma},$$
$$V_{OV1} = \frac{2}{2(1+\eta)/V_{OV2} + \lambda_\Sigma},$$
$$\frac{\beta_1}{\beta_2} = \left(1 + \eta + \frac{\lambda_\Sigma V_{OV2}}{2}\right)^2.$$

Thus M1 and M2 generally require unequal sizes even with matched PMOS process parameters. A numerical correction requires γ_p and ϕ_F, which are absent from the parameter table. The PDK operating-point values of g_m, g_{mb}, g_{ds}, and threshold voltage provide the corresponding refinement. M2's body connection remains at V_{DD} in this design.

(d) CMOS inverter with self-biased load (Fig. 7-9).

M1 is the NMOS driver. M2 is the diode-connected PMOS load, with source and body at V_{DD} and drain/gate at V_o. Both active source-body voltages are zero. At $V_{o,Q} = 0$, the load overdrive is $V_{OV2} = 0.4932\,\mathrm{V}$. The selected lengths are $L_1 = L_2 = 0.540\,\mu\mathrm{m}$, for which the assumed scaling gives

$$\lambda_\Sigma = (0.1624 + 0.1945)\frac{0.270}{0.540} = 0.17845\,\mathrm{V}^{-1}.$$

The same $C_m = 0.010\,\mathrm{pF}$ and $G_m = 318.387\,\mu\mathrm{S}$ as in part (c) apply because the selected A_0, R_S, and C_L are the same. Unity gain requires $g_{m1} = g_{m2} + g_{o1} + g_{o2}$:

$$I_D = \frac{318.387 \times 10^{-6}}{2/0.4932 + 0.17845} = 75.205\,\mu\mathrm{A},$$
$$V_{OV1} = \frac{2I_D}{G_m} = 0.472411\,\mathrm{V},$$
$$V_{in,Q} = V_{SS} + V_{TN} + V_{OV1} = 0.005411\,\mathrm{V}.$$

CMOS self-biased inverter: dimensions and response.

The continuous analytical dimensions are

Device	w_{ana} (μm)	L_{ana} (μm)	n_f	m	W_{eff} (μm)
M1 (NFET driver)	0.42201	0.54000	2	2	1.68803
M2 (PFET load)	1.75481	0.54000	2	2	7.01925

Here $g_{m2} = 304.967\,\mu\text{S}$ and $g_{o1} + g_{o2} = 13.420\,\mu\text{S}$. At $V_{o,Q} = 0$, $V_{DS1} = V_{SD2} = 0.9\,\text{V}$, so both devices are in saturation in the analytical model. The small-signal estimates are

$$\begin{aligned} A_{v0} &= -1 \quad (0\,\text{dB}), \qquad R_{\text{out}} = 3.14083\,\text{k}\Omega, \\ f_{\text{BW}} &= \text{GBW} = 2.000\,\text{MHz}, \\ f_{p1} &= 2.00031\,\text{MHz}, \qquad f_{p2} = 161.272\,\text{MHz}, \\ f_z &= 5067.29\,\text{MHz}, \qquad \phi_{\text{lag}}(f_{\text{BW}}) = 45.73^\circ. \end{aligned}$$

The calculated frequency response is the same as part (c) because the four generic-model quantities G_m, R_B, C_B, and C_C have the same selected values. Device capacitances will generally make the two responses different.

Performance with continuous analytical dimensions.

Quantity	Mirror load	Digital CMOS	PMOS load	CMOS load
I_D (μA)	26.581	120.000	71.642	75.205
$V_{in,Q}$ (V)	−0.28979	0	0.04317	0.00541
C_m (pF)	0.71683	0.75467	0.01000	0.01000
$\lvert A_{v0}\rvert$ (dB)	30.000	30.000	0.000	0.000
f_{BW} (MHz)	0.06325	0.06325	2.000	2.000
GBW (MHz)	2.000	2.000	2.000	2.000
UGF (MHz)	1.99848	1.99895	—	—
Lag at UGF (°)	92.07	89.41	—	—
Lag at f_{BW} (°)	45.12	45.04	45.73	45.73

The currents above are branch currents. The mirror-loaded inverter also draws I_{bias} in its reference branch. The summary uses unrounded dimensions and the stated approximate model, including omission of PMOS load body effect in part (c).

Grid-constrained initial dimensions.

The following initial dimensions satisfy the design grid and replace the continuous analytical values. Both w and L are selected on the 0.01 μm grid before simulation. All listed instances initially have $n_f = m = 2$, with $W_{\text{eff}} = 4w$. These initial counts remain adjustable.

Circuit	MOS	w (μm)	L (μm)	n_f	m	W_{eff} (μm)
Mirror load	M1	0.53	0.27	2	2	2.12
	M2, M3, each	2.40	0.27	2	2	9.60
Digital CMOS	M1	0.47	0.37	2	2	1.88
	M2	1.90	0.37	2	2	7.60
PMOS buffer	M1	1.00	0.27	2	2	4.00
	M2	0.84	0.27	2	2	3.36
CMOS buffer	M1	0.42	0.54	2	2	1.68
	M2	1.75	0.54	2	2	7.00

The operating points and small-signal parameters below correspond to these exact dimensions. The mirror reference current is $I_{\text{bias}} = 26.5812\,\mu$A. For the other circuits, the input bias is obtained by imposing $V_{o,Q} = 0$ on the grid-constrained dimensions. Consequently, the digital CMOS pair no longer has exactly zero input bias.

Quantity	Mirror load	Digital CMOS	PMOS buffer	CMOS buffer
I_D (μA)	26.5812	119.1389	71.9987	74.9987
$V_{in,Q}$ (V)	−0.289789	−0.000621	0.041175	0.005889
C_m (pF)	0.716827	0.754832	0.010000	0.010000
$\lvert A_{v0}\rvert$ (dB)	29.9998	30.1075	−0.0384	−0.0088
f_{BW} (MHz)	0.063247	0.062468	2.009943	1.994614
GBW (MHz)	2.000000	2.000000	2.001070	1.992597
UGF (MHz)	1.998480	1.998972	—	—
Lag at UGF (°)	92.065	89.444	—	—
Lag at f_{BW} (°)	45.123	45.039	45.732	45.727

These estimates retain the same model limitations: omitted body effect and parasitic capacitances, and an unverified $\lambda(L)$ approximation. For the grid-constrained geometry, the high-gain compensation values follow from Eq. (1) and are approximately 0.716826684 and 0.754832189 pF. The residual gain and GBW offsets follow from the selected grid-constrained dimensions. Compliance with the specifications requires simulation with the corresponding PDK devices and realizable capacitor values.

Dummy devices and physical implementation. Each active MOS has one separately named dummy instance placed beside it. Every dummy's D/G/S/B terminals connect to V_{SS}, including the PFET dummies. Dummy n_f and m remain configurable. These devices are not electrically paralleled with the active MOS, so their widths are excluded from W_{eff} in the tables. A PFET dummy requires its own V_{SS} well/tap separated from the active PFET's V_{DD} well. Each dummy retains its own integer n_f and m and legal per-finger width; the initial dummy geometry follows its associated active MOS. The physical placement must satisfy the PDK well-spacing and isolation rules.

Matching and common-centroid placement. The equal-size PMOS mirror M2/M3 uses identical units in a 2×2 AB/BA common-centroid arrangement, preserving equal centroids and consistent orientation. The unequal-width PMOS buffer uses equal-centroid interleaving of its driver and load units; this is not identical-unit matching and does not remove the devices' intentional width difference or the load's body effect. NMOS/PMOS pairs use separate well/body regions and symmetric blocks associated with each device; no cross-type common-centroid equivalence is assumed. Dummies remain adjacent to their associated active units and are not consolidated into a separate array.

Multiplicity and extracted-device correspondence. The PDK `mSwitch` convention must agree with the schematic's width, finger-count, and multiplicity interpretation. The initial schematic uses `mSwitch=1`. Its correspondence to grouped extraction for $n_f > 1$ requires CDF evaluation and comparison of schematic and extracted widths and multiplicities. Decomposition into units for common-centroid placement must preserve these same electrical quantities.

Consistency of simulated and physical dimensions. DC and AC verification use the exact grid-constrained geometry, selected PDK passive values, and intended n_f, m, and `mSwitch` settings. The schematic and layout retain these values after the specifications are met. Any subsequent change in dimensions or effective multiplicity requires renewed circuit simulation.

Model limitations and circuit verification.
The first-order model determines the initial dimensions, bias voltages, and compensation values. The PMOS-only load requires the body-effect correction above; the longer-channel CMOS designs depend on the accuracy of the assumed $\lambda(L)$ behavior. Gate capacitance, finite mirror reference impedance, junction capacitance, and physical passive models may shift both gain and bandwidth.

The DC transfer characteristic is obtained by sweeping V_{in} from -0.9 to $+0.9$ V. The two intercepts and the input bias producing $V_{out} = 0$ characterize the operating point. AC analysis about that bias gives the low-frequency gain, relative -3 dB bandwidth, gain-bandwidth product, and any positive-frequency unity-gain crossing. Dynamic phase lag excludes the DC inversion and is evaluated at UGF for the high-gain designs and at the bandwidth for the unity-gain designs. Each sizing change and parasitic extraction requires a new bias calculation.

Pre-layout simulation uses legal, grid-constrained MOS dimensions and PDK passive values. The schematic and layout share the same device parameters and connectivity, including R_S and C_m. Any dimensional change requires renewed DC and AC analysis. DRC and LVS establish geometrical legality and schematic/layout correspondence; simulation with extracted parasitic capacitances then determines the resulting circuit performance. The analytical estimates alone do not establish compliance with these requirements.

Sources.

[1] S. Palermo, *ECEN 474/704 Lab 7: Frequency Response of Inverting Amplifiers*, course laboratory handout, Figs. 7-2, 7-5, 7-6, 7-8, and 7-9; Prelab Questions 1–3.

[2] Anonymous contributor, *ECEN 474/704 Lab 1: Introduction to Cadence and MOS Device Characterization*, laboratory report, Table I: single-finger $5/0.27\ \mu$m NFET/PFET parameter measurements.

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ECEN 474/704
Lab 8: Prelab
Anonymous contributor

1. **Three-current-mirror OTA design (Fig. 8-4).** The symmetrical OTA is required to meet Table 8-1 of the lab handout:

Parameter	Requirement
Transconductance, G_m	$> 700\,\mu$S
Slew rate	$> 8\,\text{V}/\mu\text{s}$
Peak-to-peak output swing	$> 1.2\,$V
Load capacitance	$20\,$pF
Power, excluding the bias-reference branch	$< 500\,\mu$W
Power supplies	$V_{DD} = 0.9\,\text{V}, V_{SS} = -0.9\,\text{V}$

Solution.

Device parameters and model assumptions. The device parameters are taken from the single-finger $W_{\text{total}}/L = 5/0.27\,\mu$m rows of Table I in the Lab 1 report:

Parameter	NFET	PFET
$K_P = \mu C_{ox}$ ($\mu\text{A/V}^2$)	215.6	47.57
$\lvert V_{TH}\rvert$ (V)	0.4330	0.4068
λ (V^{-1})	0.1624	0.1945
θ (V^{-1})	0.9999	0.6558

All devices have $L = 0.270\,\mu$m in the initial design. The DC dimensions follow the square-law current model, with body effect, mobility degradation, and DC channel-length modulation neglected. The parameter λ enters only through the output-resistance estimate $r_o \approx 1/(\lambda I_D)$. The measured parameters depend on geometry and bias, so the resulting dimensions define the initial analytical design.

All transistor widths below are *total effective active widths*. In the Lab 1 table, W_{total} already includes all fingers. For equal parallel copies,

$$W_{\text{eff}} = m\, n_f\, w_{\text{finger}}.$$

The total width therefore already accounts for the finger count. The same relation determines the physical finger and multiplicity decomposition under the PDK width convention.

(a) Mirror ratio, current budget, and slew rate. The PMOS mirror ratio and tail current are defined as

$$N = \frac{(W/L)_{5,6}}{(W/L)_{3,4}}, \qquad I_T = I_{\text{tail}}.$$

The handout gives

$$\text{SR} \approx \frac{NI_T}{C_L}, \qquad I_{DD,\text{OTA}} \approx (N+1)I_T.$$

The bias-reference branch is excluded from the latter expression, as in the handout. The slew-rate and power limits require

$$NI_T > (20 \times 10^{-12})(8 \times 10^6) = 160\,\mu\text{A},$$
$$(N+1)I_T < \frac{500\,\mu\text{W}}{1.8\,\text{V}} = 277.778\,\mu\text{A}.$$

Hence the feasible current interval is

$$\frac{160}{N}\,\mu\text{A} < I_T < \frac{277.778}{N+1}\,\mu\text{A}.$$

It exists only when

$$160(N+1) < 277.778N \quad \Longrightarrow \quad N > 1.3585.$$

A 1:1 PMOS mirror cannot meet both constraints in this model. For $\boxed{N = 3}$, the allowed tail-current interval is $53.333\,\mu\text{A} < I_T < 69.444\,\mu\text{A}$. The selected current is

$$\boxed{I_T = I_{\text{bias}} = 60\,\mu\text{A}}$$

with a 1:1 M9–M10 bias mirror. Then

$$\text{SR} \approx \frac{3(60\,\mu\text{A})}{20\,\text{pF}} = \boxed{9\,\text{V}/\mu\text{s}},$$
$$P_{\text{OTA}} \approx (1.8\,\text{V})(4)(60\,\mu\text{A}) = \boxed{432\,\mu\text{W}}.$$

The ideal current model predicts constant OTA power within the saturation-valid input range. Finite output conductance introduces current variation with common-mode voltage; the DC sweep determines the maximum power over this range. Including the separate $60\,\mu\text{A}$ bias-reference branch gives approximately $540\,\mu\text{W}$. Table 8-1 applies to the OTA power with this branch excluded.

(b) Input transconductance and overdrive. At balanced input, $I_{D1} = I_{D2} = I_T/2 = 30\,\mu\text{A}$ and $G_m = Ng_{m1}$. The minimum input-device transconductance is

$$g_{m1} > \frac{700\,\mu\text{S}}{3} = 233.333\,\mu\text{S}.$$

With $V_{OV1,2} = 0.200\,\text{V}$, the transconductances are

$$g_{m1} = \frac{2(30\,\mu\text{A})}{0.200\,\text{V}} = 300\,\mu\text{S}, \qquad \boxed{G_m = 900\,\mu\text{S} > 700\,\mu\text{S}.}$$

(c) Transistor dimensions and matching. The selected overdrives are $V_{OV} = 0.200\,\mathrm{V}$ for M1–M8 and $V_{OV} = 0.150\,\mathrm{V}$ for the bias/tail devices M9 and M10. At the balanced operating point,

$$\begin{aligned} I_{D1,2,3,4} &= I_T/2 = 30\,\mu\mathrm{A}, \\ I_{D5,6,7,8} &= NI_T/2 = 90\,\mu\mathrm{A}, \\ I_{D9,10} &= 60\,\mu\mathrm{A}. \end{aligned}$$

The square-law current relation gives

$$I_D \approx \frac{1}{2}K_P\frac{W}{L}V_{OV}^2, \qquad \frac{W}{L} = \frac{2I_D}{K_P V_{OV}^2}, \qquad W = L\frac{W}{L}.$$

Using currents in μA and K_P in $\mu\mathrm{A/V}^2$,

$$\begin{aligned} (W/L)_{1,2} &= \frac{2(30)}{215.6(0.200)^2} = 6.9573, \\ (W/L)_{3,4} &= \frac{2(30)}{47.57(0.200)^2} = 31.5325, \\ (W/L)_{5,6} &= \frac{2(90)}{47.57(0.200)^2} = 94.5974, \\ (W/L)_{7,8} &= \frac{2(90)}{215.6(0.200)^2} = 20.8720, \\ (W/L)_{9,10} &= \frac{2(60)}{215.6(0.150)^2} = 24.7372. \end{aligned}$$

Devices	Type	I_D (μA)	V_{OV} (V)	W (μm)	L (μm)
M1, M2	NFET	30	0.200	1.8785	0.270
M3, M4	PFET	30	0.200	8.5138	0.270
M5, M6	PFET	90	0.200	25.5413	0.270
M7, M8	NFET	90	0.200	5.6354	0.270
M9, M10	NFET	60	0.150	6.6790	0.270

The displayed widths are analytical values before grid rounding. The equal-length PMOS dimensions give

$$\frac{W_{5,6}}{W_{3,4}} = 3.$$

A physical implementation with identical unit devices preserves this ratio. The matched pairs are M1/M2, M3/M4, M5/M6, M7/M8, and M9/M10. The M3–M5 and M4–M6 mirrors use consistent unit geometry with the intended 1:3 ratio. NMOS bodies connect to V_{SS}; PMOS bodies connect to V_{DD}.

Dummy MOS devices do not contribute to the active widths above. NFET dummy terminals connect to V_{SS} and PFET dummy terminals to V_{DD}. Consistent dummy multiplicity in the schematic and layout allows LVS to compare the same devices under the PDK convention.

(d) DC bias and input common-mode range. The quiescent input is $V_{CM,Q} = 0.300\,\mathrm{V}$ with zero differential input. The approximate first-stage and mirror-reference voltages are

$$\begin{aligned} V_{S1} = V_{S2} &= 0.300 - (0.4330 + 0.200) = -0.3330\,\mathrm{V}, \\ V_{A1} = V_{A2} &= 0.900 - (0.4068 + 0.200) = 0.2932\,\mathrm{V}, \\ V_B &= -0.900 + 0.4330 + 0.200 = -0.2670\,\mathrm{V}, \\ V_{G9} = V_{G10} &= -0.900 + 0.4330 + 0.150 = -0.3170\,\mathrm{V}. \end{aligned}$$

The reference $V_{o,Q} \approx V_B$ gives equal drain voltages within the matched M5/M6 and M7/M8 pairs. Finite output conductance determines the actual open-loop output bias; the DC model without channel-length modulation does not uniquely fix it. Setting $V_{o,Q} = 0$ therefore requires a suitable differential input offset or feedback arrangement.

At the reference point $V_{o,Q} = -0.2670\,\mathrm{V}$, the saturation checks are

Devices	V_{DS} or V_{SD} (V)	V_{OV} (V)	Margin (V)
M1, M2	0.6262	0.200	0.4262
M3, M4	0.6068	0.200	0.4068
M5, M6	1.1670	0.200	0.9670
M7, M8	0.6330	0.200	0.4330
M9	0.5830	0.150	0.4330
M10	0.5670	0.150	0.4170

All active devices are saturated in the assumed model. The margin is $V_{DS} - V_{OV}$ for an NFET or $V_{SD} - V_{OV}$ for a PFET.

Keeping the input devices and tail source saturated gives

$$\begin{aligned} V_{CM,\mathrm{min}} &\approx V_{SS} + V_{OV10} + V_{TN1} + V_{OV1} \\ &= -0.900 + 0.150 + 0.4330 + 0.200 = -0.1170\,\mathrm{V}, \\ V_{CM,\mathrm{max}} &\approx V_{DD} - |V_{TP3}| - V_{OV3} + V_{TN1} \\ &= 0.900 - 0.4068 - 0.200 + 0.4330 = 0.7262\,\mathrm{V}. \end{aligned}$$

Thus the estimated interval is $-0.1170 < V_{CM} < 0.7262\,\mathrm{V}$, with width $0.8432\,\mathrm{V}$. These bounds neglect body effect. The transistor operating regions in the common-mode DC sweep define the actual input range.

(e) Saturation-based output swing. M6 and M8 set the output-node compliance:

$$V_{o,\max} \approx V_{DD} - V_{OV6} = 0.900 - 0.200 = 0.700\,\text{V},$$
$$V_{o,\min} \approx V_{SS} + V_{OV8} = -0.900 + 0.200 = -0.700\,\text{V}.$$

The estimated saturation-valid interval is therefore

$$\boxed{\Delta V_o = V_{o,\max} - V_{o,\min} = 1.400\,\text{V} > 1.2\,\text{V}.}$$

In the handout's DC-sweep method, the endpoints occur where either output device leaves saturation, and their difference gives the output-swing interval. The maximum symmetric waveform also depends on the quiescent output voltage. For example, about $V_{o,Q} = -0.2670\,\text{V}$,

$$v_{o,\text{pk}} < \min(0.4330, 0.9670)\,\text{V} = 0.4330\,\text{V}.$$

A waveform centered near zero therefore requires bias or feedback that places the quiescent output near zero.

(f) Output resistance, gain, and frequency estimates. Using the Lab 1 λ values at the initial length,

$$r_{o6} \approx \frac{1}{0.1945(90\times10^{-6})} = 57.127\,\text{k}\Omega,$$
$$r_{o8} \approx \frac{1}{0.1624(90\times10^{-6})} = 68.418\,\text{k}\Omega,$$
$$R_{\text{out}} \approx r_{o6} \parallel r_{o8} = 31.132\,\text{k}\Omega,$$
$$A_{v0} \approx G_m R_{\text{out}} = 28.019\ \text{V/V} \quad (28.95\,\text{dB}).$$

With the 20 pF load assumed to dominate output capacitance,

$$f_{p1} \approx \frac{1}{2\pi R_{\text{out}} C_L} = 255.61\,\text{kHz},$$
$$\text{GBW} \approx A_{v0} f_{p1} = \frac{G_m}{2\pi C_L} = 7.162\,\text{MHz}.$$

The factor 2π converts the handout's angular frequency to hertz. The output-pole approximation omits internal-node poles, which also affect the unity-gain frequency and phase margin. The unity-feedback `iprobe/stb` configuration determines the phase margin of the transistor circuit. Table 8-1 specifies no numerical gain, GBW, or phase-margin limit.

(g) Design summary.

Quantity	Requirement	First-order estimate
Transconductance	$> 700\,\mu$S	$900\,\mu$S
Slew rate	$> 8\,\mathrm{V}/\mu\mathrm{s}$	$9\,\mathrm{V}/\mu\mathrm{s}$
Saturation-based output interval	$> 1.2\,$V	$1.4\,$V
OTA power, excluding bias branch	$< 500\,\mu$W	$432\,\mu$W
Load capacitance	$20\,$pF	$20\,$pF
Supply rails	$\pm 0.9\,$V	$\pm 0.9\,$V
Low-frequency voltage gain	Characterize	28.95 dB
Dominant pole	Characterize	255.61 kHz
Gain-bandwidth product	Characterize	7.162 MHz
Phase margin	Characterize	Requires `stb`

The calculated transconductance, slew rate, output interval, and OTA power satisfy the specified limits within the stated model. Power follows the ideal current budget, while slew rate assumes fully steered current. Mirror dynamics and parasitic capacitances affect the actual rising and falling slopes.

The DC operating points establish the saturation-valid input range. The maximum OTA power is evaluated over this range, excluding the reference branch specified by the handout. The output-device region transitions in a differential DC sweep determine the output-swing limits.

For AC characterization, the measured transconductance is the ratio of the output current flowing into C_L to the differential input voltage. The first-order relation is

$$\frac{I_{C_L}(s)}{V_{id}(s)} \approx G_m \frac{sR_{\mathrm{out}}C_L}{1 + sR_{\mathrm{out}}C_L}.$$

This current ratio tends to zero at DC, while internal poles produce the high-frequency rolloff. The $700\,\mu$S specification therefore applies to the passband maximum. An input-offset correction establishes an active-region output bias for the open-loop measurement. Rising and falling slew rates and the unity-buffer phase margin complete the dynamic characterization; parasitic extraction includes the effect of the physical layout.

The Lab 6 report provides a comparison between a first-order calculation and transistor-level simulation: the simple amplifier's pre-layout gain is 24.0950 dB, compared with the calculated 27.01 dB. The OTA values above are analytical predictions under the stated device-model assumptions.

Sources.

[1] S. Palermo, *ECEN 474/704 Lab 8: Operational Transconductance Amplifiers*, Fig. 8-4, Table 8-1, and the output-swing and stability-analysis instructions.

[2] Anonymous contributor, *ECEN 704 Lab 1 Report*, Table I: single-finger $W_{\mathrm{total}}/L = 5/0.27\,\mu$m parameter rows.

[3] Anonymous contributor, *ECEN 704 Lab 6 Report*, Tables 1–2.

/November 6, 2026

ECEN 474/704
Lab 9: Prelab
Anonymous contributor

1. **Two-stage Miller operational-amplifier design (Fig. 9-3).** Design the amplifier to meet Table 9-1 of the laboratory handout.

Parameter	Requirement
Open-loop DC voltage gain, A_{v0}	$> 55\,\text{dB}$
Common-mode rejection ratio	$> 60\,\text{dB}$
Gain–bandwidth product	$> 3\,\text{MHz}$
Phase margin in unity-gain feedback	$> 45^\circ$
Peak-to-peak output swing	$> 1.2\,\text{V}$
Load capacitance, C_L	$30\,\text{pF}$
Load resistance, R_L	∞ (open circuit)
Power, including the bias-reference branch	$< 500\,\mu\text{W}$
Power supplies	$V_{DD} = +0.9\,\text{V}, \quad V_{SS} = -0.9\,\text{V}$

Solution.

Device parameters and assumptions. The device parameters are taken from the single-finger $W_{\text{total}} = 5\,\mu\text{m}$, $L = 0.27\,\mu\text{m}$ measurements in Table I of the Lab 1 report:

Parameter	NFET	PFET
$K_P = \mu C_{ox}$ ($\mu\text{A/V}^2$)	215.6	47.57
$\lvert V_{TH}\rvert$ (V)	0.4330	0.4068
λ (V^{-1})	0.1624	0.1945
θ (V^{-1})	0.9999	0.6558

All transistors have $\boxed{L = 0.270\,\mu\text{m}}$. The square-law model gives

$$I_D \approx \frac{1}{2} K_P \frac{W_{\text{eff}}}{L} V_{OV}^2, \qquad g_m \approx \frac{2I_D}{V_{OV}}, \qquad r_o \approx \frac{1}{\lambda I_D}.$$

The DC analysis neglects body effect, mobility degradation, and channel-length modulation. Thus, the listed θ does not enter the sizing equations. Finite r_o is retained in the small-signal model, while intrinsic node capacitances are neglected in the initial frequency analysis. The results below follow from these approximations.

(a) Current budget and transconductance selection.

In Fig. 9-3, M1/M2 form the NMOS input pair, M3/M4 the PMOS mirror load, M5 the diode-connected NMOS bias reference, M6 the tail source, M7 the output sink, and M8 the PMOS gain device. C_c connects the first-stage output to the final output.

Including the reference branch, the power limit imposes

$$I_{\text{total}} < \frac{500\,\mu\text{W}}{V_{DD} - V_{SS}} = \frac{500\,\mu\text{W}}{1.8\,\text{V}} = 277.778\,\mu\text{A}.$$

The selected bias currents and compensation capacitance are

$$\boxed{I_{\text{bias}} = 10\,\mu\text{A}, \quad I_T = 40\,\mu\text{A}, \quad I_2 = 160\,\mu\text{A}, \quad C_c = 10\,\text{pF}.}$$

At balanced input, $I_{D1,2,3,4} = 20\,\mu\text{A}$. The ideal current ratios are M6:M5 = 4 : 1 and M7:M5 = 16 : 1, giving

$$P_Q = (1.8\,\text{V})(10 + 40 + 160)\,\mu\text{A} = \boxed{378\,\mu\text{W} < 500\,\mu\text{W}}.$$

The nominal power margin is therefore $122\,\mu\text{W}$. A DC common-mode sweep determines the maximum power over the valid input range.

From GBW $\approx g_{m1}/(2\pi C_c)$,

$$g_{m1} > 2\pi(3\,\text{MHz})(10\,\text{pF}) = 188.50\,\mu\text{S}.$$

An input overdrive of $V_{OV1,2} = 0.160\,\text{V}$ gives $g_{m1,2} = 250\,\mu\text{S}$. For the remaining devices, the selected overdrives are $V_{OV3,4,8} = 0.200\,\text{V}$ and $V_{OV5,6,7} = 0.150\,\text{V}$, yielding

$$g_{m3,4} = 200\,\mu\text{S}, \qquad g_{m8} = 1.600\,\text{mS}.$$

The resulting estimate is $g_{m1}/(2\pi C_c) = 3.979\,\text{MHz}$. Part (f) includes the second pole and the feedforward zero in the unity-gain frequency and phase-margin calculation.

(b) Continuous analytical dimensions. Solving the drain-current equation for the aspect ratio, with currents in μA and K_P in $\mu\text{A/V}^2$, gives

$$\frac{W_{\text{eff}}}{L} = \frac{2I_D}{K_P V_{OV}^2}.$$

For example, $(W/L)_{1,2} = 40/[215.6(0.16)^2] = 7.24722$. The resulting continuous widths are:

Devices	Type	I_D (μA)	V_{OV} (V)	W_{eff} (μm)	L (μm)
M1, M2	NFET	20	0.160	1.95675	0.270
M3, M4	PFET	20	0.200	5.67585	0.270
M5	NFET	10	0.150	1.11317	0.270
M6	NFET	40	0.150	4.45269	0.270
M7	NFET	160	0.150	17.81076	0.270
M8	PFET	160	0.200	45.40677	0.270

(c) Grid-constrained initial sizes and dummy arrangement.

The analytical dimensions are rounded to the 0.01 μm design grid. For a finger width w, a finger count n_f, and a parallel multiplicity m, the effective width is

$$W_{\text{eff}} = w\, n_f\, m, \qquad w_t = w\, n_f.$$

The instance width w_t excludes m. The entries below apply to each named transistor individually.

Device	w (μm)	L (μm)	n_f	m	W_{eff} (μm)
M1, M2	0.98	0.27	1	2	1.96
M3, M4	1.42	0.27	1	4	5.68
M5	1.11	0.27	1	1	1.11
M6	1.11	0.27	1	4	4.44
M7	1.11	0.27	1	16	17.76
M8	1.42	0.27	1	32	45.44

These dimensions preserve the NMOS bias ratios exactly and give $(W/L)_8/(W/L)_3 = 8$. Hence, the 160 μA output PMOS and 20 μA mirror devices have equal square-law overdrive. The gate of M8 is driven by the first-stage output and is separate from the diode-connected gate of M3.

For $I_{\text{bias}} = 10\,\mu$A, the rounded dimensions give

$$V_{OV1,2} = 0.159867\,\text{V}, \quad V_{OV3,4,8} = 0.199927\,\text{V}, \quad V_{OV5,6,7} = 0.150214\,\text{V}.$$

Thus $g_{m1,2} = 250.208\,\mu$S and $g_{m8} = 1.600585\,$mS. The subsequent calculations use these dimensions.

Matching and dummies. M1/M2 use equal units in an AB/BA common-centroid arrangement. M3/M4 likewise use equal unit widths, equal centroids, and symmetric routing. M5/M6/M7 use the same unit size and orientation; their different multiplicities set the intended current ratios. The resulting arrangement contains 65 active one-finger units.

Each active MOS symbol has one adjacent dummy *symbol* with the same unit w and L. A dummy multiplicity of $n_f = 1$, $m = 2$ represents two physical edge units per active device group (16 dummy units in total). NFET dummy D/G/S/B all connect to V_{SS}; PFET dummy D/G/S/B all connect to V_{DD}. The physical edge units follow the boundaries of the matched groups. Since the dummies carry no signal current, their widths are excluded from active W_{eff}.

Consistent schematic and extracted netlists require the same finger widths, finger counts, and multiplicities. This includes the CDF width convention and the extraction `mSwitch` setting. The dimensions above define the initial design; pre-layout DC and AC simulations determine the final dimensions.

(d) Operating point, common-mode range, and output headroom.

The selected input bias is $V_{CM,Q} = 0.150\,\text{V}$, with a nominal output target of $V_{o,Q} = 0$. Let v_x be the M1/M3 drain node, v_y the M2/M4 drain and M8 gate node, and v_s the input-pair source node. Substitution of the rounded dimensions gives

$$\begin{aligned} V_B &= V_{SS} + V_{TN} + V_{OV5} = -0.316786\,\text{V}, \\ V_s &= V_{CM,Q} - V_{TN} - V_{OV1} = -0.442867\,\text{V}, \\ V_x &= V_{DD} - |V_{TP}| - V_{OV3} = 0.293273\,\text{V}, \\ V_y &= V_{DD} - |V_{TP}| - V_{OV8} = 0.293273\,\text{V}. \end{aligned}$$

At this output voltage, $V_{DS7} = V_{SD8} = 0.9\,\text{V}$. Also, $V_{DS1,2} = 0.736140\,\text{V}$, $V_{SD3,4} = 0.606727\,\text{V}$, and $V_{DS6} = 0.457133\,\text{V}$, all above their selected overdrives. With channel-length modulation neglected, equal output-branch currents do not uniquely determine $V_{o,Q}$. An input-offset sweep establishes the bias that produces $V_o = 0$ for open-loop measurements.

The input-pair and tail-source saturation conditions give

$$\begin{aligned} V_{CM,\text{min}} &\approx V_{SS} + V_{OV6} + V_{TN} + V_{OV1} = -0.156919\,\text{V}, \\ V_{CM,\text{max}} &\approx V_{DD} - |V_{TP}| - V_{OV3} + V_{TN} = 0.726273\,\text{V}. \end{aligned}$$

The selected 0.150 V lies inside this interval. These bounds follow from the stated threshold-voltage approximation; the transistor-level DC sweep includes body effect and the associated bias shifts.

At the output, the saturation-limited window is

$$V_{SS} + V_{OV7} < V_o < V_{DD} - V_{OV8}, \qquad -0.749786 < V_o < 0.700073\,\text{V}.$$

Its total span is 1.449859 V, and the symmetric headroom about zero is approximately $\boxed{1.400146\,\text{V}_{pp} > 1.2\,\text{V}_{pp}}$. This interval is set by transistor saturation. The swing at a specified distortion is obtained from 100 Hz open-loop PSS at the input amplitude corresponding to 5% THD.

Slew rate. The tail current limits the rate of change of the compensation capacitor voltage, giving

$$\text{SR} \approx I_T/C_c = \boxed{4.0\,\text{V}/\mu\text{s}}.$$

For falling output, M7 must sink both compensation and load current:

$$\frac{I_2 - I_T}{C_L} = \frac{160 - 40}{30}\frac{\mu\text{A}}{\text{pF}} = 4.0\,\text{V}/\mu\text{s}.$$

A rising $4\,\text{V}/\mu\text{s}$ output requires about $I_8 = I_7 + I_T + C_L\text{SR} = 320\,\mu\text{A}$, twice its quiescent current. The rising slew-rate estimate therefore requires M8 to supply this current. Transient analysis determines the rising and falling slew rates; Lab 9 specifies no numerical minimum.

(e) Finite-output-resistance gain and CMRR.

The common-mode response depends on the finite tail-source conductance. For the small-signal analysis, define

$$n = g_{m1},\ p = g_{m3},\ g = g_{ds1},\ q = g_{ds3},\ t = g_{ds6}, \quad u = g + q, \quad h = n + g.$$

Matched pairs have equal small-signal parameters. With the supplies and bias gate at AC ground, the first-stage nodal matrix is

$$\begin{bmatrix} u+p & 0 & -h \\ p & u & -h \\ -g & -g & t+2h \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_s \end{bmatrix} = \begin{cases} [n/2,\ -n/2,\ 0]^T v_{id}, \\ [-n,\ -n,\ 2n]^T v_{cm}. \end{cases}$$

Here $v_{id} = v_{i+} - v_{i-}$, with M2 receiving $+v_{id}/2$. Let $D = t(u+p) + 2h(p+q)$. Solving gives

$$A_{d1} = \frac{v_y}{v_{id}} = -\frac{n[t(u+2p) + 2h(2p+q)]}{2uD},$$
$$A_{cm1} = \frac{v_y}{v_{cm}} = -\frac{nt}{D}.$$

For the grid sizes, in μS,

$$n = 250.2076, \quad p = 200.0732, \quad g = 3.248, \quad q = 3.890, \quad t = 6.496.$$

Therefore

$$A_{d1} = -34.71517, \qquad A_{cm1} = -0.01551834,$$

$$\boxed{\text{CMRR} = 20\log_{10}\left|\frac{A_{d1}}{A_{cm1}}\right| = 66.9935\,\text{dB} > 60\,\text{dB}.}$$

The second-stage gain cancels in this ratio. The result retains finite tail r_o under the assumptions of matched devices and negligible body effect.

The first-stage output resistance is found by setting both input voltages to zero and injecting a test current into v_y:

$$R_A = \frac{(u+p)(t+2h) - gh}{uD} = 141.1964\,\text{k}\Omega.$$

Thus its equivalent differential transconductance is $G_{mA} = |A_{d1}|/R_A = 245.8644\,\mu$S. For the second stage,

$$R_B = r_{o7} \parallel r_{o8} = \frac{1}{(\lambda_n + \lambda_p)I_2} = 17.5119\,\text{k}\Omega,$$
$$A_2 = -g_{m8}R_B = -28.02930,$$
$$A_{v0} = A_{d1}A_2 = 973.0421\,\text{V/V},$$
$$20\log_{10}|A_{v0}| = \boxed{59.7626\,\text{dB} > 55\,\text{dB}}.$$

(f) Miller compensation and phase margin.

The first stage is represented by its Norton equivalent from part (e), with $C_A = 0$, $C_B = C_L = 30\,\text{pF}$, and $C_c = 10\,\text{pF}$. The two-node model gives

$$H(s) = \frac{A_{v0}(1 - s/\omega_z)}{1 + as + bs^2}, \qquad \omega_z = \frac{g_{m8}}{C_c},$$

$$a = R_A(C_A + C_c) + R_B(C_B + C_c) + g_{m8}R_AR_BC_c,$$
$$b = R_AR_B(C_AC_B + C_AC_c + C_BC_c).$$

For the selected dimensions,

$$a = 4.1688808 \times 10^{-5}\,\text{s}, \qquad b = 7.4178551 \times 10^{-13}\,\text{s}^2.$$

The right-half-plane zero adds phase lag. The two poles are $s = -\omega_{p1}$ and $s = -\omega_{p2}$:

$$\omega_{p1} = \frac{2}{a + \sqrt{a^2 - 4b}}, \qquad \omega_{p2} = \frac{a + \sqrt{a^2 - 4b}}{2b}.$$

Converting angular frequencies to Hz,

$$\boxed{f_{p1} = 3.81932\,\text{kHz}, \quad f_{p2} = 8.94079\,\text{MHz}, \quad f_z = 25.47411\,\text{MHz}.}$$

Gain–bandwidth product and unity-gain frequency. The bandwidth f_{BW} satisfies $|H(j2\pi f_{\text{BW}})| = A_{v0}/\sqrt{2}$, and the gain–bandwidth product is $\text{GBW} = A_{v0}f_{\text{BW}}$. The unity-gain frequency f_u satisfies $|H(j2\pi f_u)| = 1$. With $x = \omega_u^2$, this condition reduces to

$$b^2x^2 + \left(a^2 - 2b - \frac{A_{v0}^2}{\omega_z^2}\right)x + 1 - A_{v0}^2 = 0.$$

Taking its positive solution yields

$$f_{\text{BW}} = 3.81932\,\text{kHz},$$
$$\text{GBW} = \boxed{3.71636\,\text{MHz} > 3\,\text{MHz}},$$
$$f_u = \boxed{3.49386\,\text{MHz}}.$$

For unity negative feedback,

$$\text{PM} = 180^\circ - \tan^{-1}\frac{f_u}{f_{p1}} - \tan^{-1}\frac{f_u}{f_{p2}} - \tan^{-1}\frac{f_u}{f_z} = \boxed{60.9086^\circ > 45^\circ}.$$

The two-node model therefore satisfies the phase-margin requirement. The calculation excludes intrinsic mirror and tail-node poles, body effect, and layout parasitics. Unity-feedback STB analysis evaluates their effect on the complete transistor circuit.

(g) Summary.

All values below use the grid-constrained geometry in part (c).

Quantity	Calculated value	Specification / note
DC gain	59.76 dB	> 55 dB
CMRR	66.99 dB	> 60 dB
GBW	3.716 MHz	> 3 MHz
Unity-gain frequency	3.494 MHz	Actual model crossing
Phase margin	60.91°	$> 45°$
Power including bias	378 μW	$< 500\,\mu$W
Centered output headroom	1.400 V_{pp}	$> 1.2\,V_{pp}$ target
5%-THD PSS swing	To be simulated	Not established by headroom
Common-mode input range	$[-0.157,\ 0.726]$ V	Body effect omitted
Slew-rate estimate	4.0 V/μs	No specified minimum
C_c, C_L, R_L	10 pF, 30 pF, ∞	Selected compensation/load

With $C_A = 1$ pF at the first-stage output, the two-node model gives GBW = 3.704 MHz, $f_u = 3.429$ MHz, and PM = 58.97°. Thus, the selected compensation maintains both the gain–bandwidth and phase-margin requirements with this additional node capacitance.

Within the model used here, the calculated gain, CMRR, GBW, phase margin, and quiescent power satisfy the specified limits. The dimensions in part (c), together with $C_c = 10$ pF, therefore form the initial design. The 1.400 V_{pp} output headroom exceeds the 1.2 V_{pp} swing target. This headroom follows from the saturation constraints; the output swing at 5% THD is determined by the 100 Hz PSS test.

The closed-loop IM3 test uses 10 kHz and 11 kHz tones with an amplitude of 300 mV each. The sum of the tone amplitudes is 0.6 V, greater than the available input excursion about $V_{CM} = 0.150$ V. The specified test therefore extends beyond the calculated common-mode input range. PSS determines the resulting intermodulation products, while noise, PSRR, and sensitivity to process variation and mismatch are evaluated in their respective transistor-level analyses.

References.

[1] *ECEN 474/704 Lab 9: Two-Stage Miller Operational Amplifier*, course handout: Fig. 9-3, small-signal model, and Table 9-1; PSS swing and report instructions.

[2] Anonymous contributor, *ECEN 474/704 Lab 1 Report*, Table I: single-finger $W = 5\,\mu$m, $L = 0.27\,\mu$m NFET/PFET parameter rows.